

Review Article

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Signed graphs with strong (anti-)reciprocal eigenvalue property[#]<https://doi.org/10.1515/spma-2024-0017>

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Abstract: A (signed) graph is said to exhibit the strong reciprocal (anti-reciprocal) eigenvalue property (SR) (resp., (-SR)) if for any eigenvalue λ , it has $\frac{1}{\lambda}$ (resp., $-\frac{1}{\lambda}$) as an eigenvalue as well, with the same multiplicity. It is well known that the corona of a (signed) graph does have the property -SR, and if the graph has symmetric spectrum, then it also has the property SR. Therefore, it is interesting to identify (signed) graphs which are not corona graphs with the properties SR or -SR. Recently, a few constructions for unsigned graphs with the property -SR have been offered. In this article, we extend such constructions to signed graphs.

Keywords: adjacency spectrum, corona product, Kronecker product, subdivision

MSC 2020: 05C50, 05C22

1 Introduction

In this article, we deal with graphs with signs on the edges which are called signed graphs. More formally, let $G = (V, E)$ be a simple graph with vertex set $V = \{1, 2, \dots, n\}$ and edge set E ; a signed graph $\Sigma = (G, \sigma) = G_\sigma$ is a graph $G = (V, E)$ together with a function $\sigma : E \rightarrow \{+1, -1\}$ assigning a positive or negative sign to each edge. The unsigned graph G is said to be the underlying graph of Σ , while the function σ is called the signature of Σ . As usual, the (unsigned) graphs can be thought of as signed graphs equipped with the all-positive signature $\sigma = +: E \rightarrow \{+1\}$. If two vertices i and j are adjacent in G (or Σ), we denote it by $i \sim j$ and say $\{i, j\} = ij \in E$. The adjacency matrix of $\Sigma = (G, \sigma)$ is defined as an $n \times n$ matrix $A(\Sigma) = [a_{ij}]$, where $a_{ij} = \sigma(ij)$ if $i \sim j$ in G and $a_{ij} = 0$ otherwise. Clearly since $A(\Sigma)$ is a real symmetric matrix, all of its eigenvalues are real. Let $\lambda_1(\Sigma) \leq \lambda_2(\Sigma) \leq \dots \leq \lambda_n(\Sigma)$ be the eigenvalues of $A(\Sigma)$ arranged in non-decreasing order. The characteristic polynomial of Σ is the characteristic polynomial of $A(\Sigma)$, and it is denoted by $P(\Sigma; x) = \det(xI - A(\Sigma)) = \sum_{i=0}^n a_i x^i$. A signed graph Σ is said to be nonsingular (respectively singular) if $A(\Sigma)$ is nonsingular (respectively singular). The spectrum of Σ is denoted as $\text{Spec}(\Sigma)$ and is made up of the eigenvalues of $A(\Sigma)$. When we say eigenvalues of Σ we are referring to the eigenvalues of $A(\Sigma)$.

The notation for unsigned graphs naturally extends to signed graphs so the degree of a vertex is equal to the number of edges it is incident to, regardless of their signs. However, we also have the notion of the net-degree of a vertex which is specific to signed graphs and is defined as being equal to the difference between the number of positive edges and the number of negative edges incident to the considered vertex. A signed graph

[#] Dedicated to Professor Nair Maria Abreu for her (70 + 3)rd birthday.

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is said to be net-regular if all of its vertices have the same net-degree (note that the underlying graph need not be regular in the usual sense). The sign of a cycle is the product of its edge signs, so a cycle is positive if and only if it contains an even number of negative edges. A signed graph is said to be balanced if and only if the product of the edge signs in any closed walk is positive. Hence, a signed graph is balanced if and only if all of its cycles, if any, are positive. Trees are evidently balanced signed graphs.

An important feature of signed graphs is the concept of switching the signature. Given a signed graph $\Sigma = (G, \sigma)$ and a subset $U \subseteq V(G)$, we can let Σ^U be the signed graph obtained from Σ by reversing the signs of the edges so that $\sigma_{\Sigma^U}(e) = -\sigma_{\Sigma}(e)$ for any edge e between U and $V \setminus U$ and $\sigma_{\Sigma^U}(e) = \sigma_{\Sigma}(e)$ otherwise. The signed graph Σ^U is then said to be switching equivalent to Σ . Two signed graphs Σ and Σ' on G are isomorphic if there is an automorphism on G that preserves edge signs, and this relation is denoted by $\Sigma \equiv \Sigma'$. The signed graphs Σ and Σ' are switching isomorphic if there exists a subset $U \subseteq V(G)$ such that $\Sigma \equiv \Sigma'^U$. Switching isomorphic signed graphs are naturally cospectral, so they are indistinguishable from their spectra, and in this context, they represent the same combinatorial object. It is well known that a signed graph is balanced if and only if it is switching equivalent to the all-positive signature.

It is also well known that for unsigned graphs the spectrum of a graph is symmetric with respect to the origin if and only if it is a bipartite graph. For short, we say that bipartite graphs are spectrally symmetric. It is true that a signed graph also has a symmetric spectrum if it is bipartite but there do exist spectrally symmetric signed graphs which are not bipartite. A signed graph $\Sigma = (G, \sigma)$ is said to be sign-symmetric if Σ is switching isomorphic to its negation $-\Sigma = (G, -\sigma)$. Since λ is an eigenvalue of Σ if and only if $-\lambda$ is an eigenvalue of $-\Sigma$, we immediately have that a sign-symmetric graph is spectrally symmetric. Notably, there exist spectrally symmetric signed graphs being not sign-symmetric [8,11].

We now consider unsigned graphs. A graph G is said to have the reciprocal eigenvalue property known as property (R) if for each eigenvalue λ of G its reciprocal $\frac{1}{\lambda}$ is also an eigenvalue of G . Further to this if the eigenvalues λ and $\frac{1}{\lambda}$ appear with the same multiplicity in the spectrum of G , then this property is referred to as the strong reciprocal eigenvalue property known as property (SR). Analogously, if for each λ in the spectrum of a graph G it is also true that $-\frac{1}{\lambda}$ is in the spectrum of G , then this graph is said to have the anti-reciprocal eigenvalue property known as property (-R). In the same way as before if λ and $-\frac{1}{\lambda}$ appear with the same multiplicity in the spectrum of G , then the property is referred to as the strong anti-reciprocal eigenvalue property known as property (-SR). There are several papers dealing with such properties and we refer the reader to [1,3,6, 12,13] and the references therein for some of them.

Let G and H be two graphs on disjoint sets of n and m vertices, respectively. The corona product $G \circ H$ of G and H is defined as the graph obtained by taking one copy of G and n copies of H and then joining the i th ($i = 1, 2, \dots, n$) vertex of G to every vertex in the i th copy of H . If $H = K_1$, then the graph $G \circ K_1$ is called a simple corona graph and is obtained by adding a single pendant vertex and edge to each vertex in G . For the corona product of two signed graphs, there are many different ways in which the newly introduced edges can be signed.

In [7] Barik et al. proved that if $G = F \circ K_1$ for some graph F , then G satisfies the property (-SR). Furthermore, whenever F is bipartite, it is shown that G satisfies property (SR). Note that if F is a bipartite graph, then so is G and it shall be spectrally symmetric meaning that the properties (SR) and (-SR) are equivalent for such graphs.

Let G and H be two graphs with vertex sets $V(G) = \{v_1, v_2, \dots, v_n\}$ and $V(H) = \{u_1, u_2, \dots, u_m\}$, respectively. Introduced by Weichsel in [15], the Kronecker product of G and H denoted by $G \otimes H$ is the graph with vertex set $V(G) \times V(H)$, where two vertices (v_i, u_k) and (v_j, u_l) are adjacent if and only if v_i and v_j are adjacent in G and u_k and u_l are adjacent in H .

The singular value decomposition of an $m \times n$ complex matrix M is a factorisation of the form $M = U\Sigma V^*$, where U is an $m \times m$ complex unitary matrix and Σ is an $m \times n$ rectangular diagonal matrix with non-negative real numbers along its diagonal and V is an $n \times n$ complex unitary matrix with V^* being its conjugate transpose. The diagonal entries $\sigma_i = \Sigma_{ii}$ of Σ are uniquely determined by M and are known as the singular values of M . The columns of U and the columns of V are called left-singular vectors and right-singular vectors of M , respectively. Each pair of singular vectors u_i and v_i corresponds to the singular value σ_i .

Note that the concept of reciprocal and anti-reciprocal spectra is defined for unsigned graphs, but it can also be considered for signed graphs in the natural way. Similarly, the corona of two signed graphs can be considered where all newly introduced connecting edges are assigned to be positive. The purpose of this article is to identify classes of signed graphs satisfying properties (SR) or (-SR), and this is done by generalising to signed graphs the results obtained for unsigned graphs. Most of these results are inspired by the recent paper [14]. In many cases, this can be done with a suitable modification, especially when regularity is requested. However, some further families with no regularity do exhibit such properties but we could not prove them. In this event, such cases will be offered as open problems.

Here is the remainder of the article. In Section 2, we collect some basic facts and results on properties (SR) and (-SR) for signed graphs, in particular, we show that the simple corona of a signed graph with a symmetric spectrum leads to a signed graph with both the (SR) and (-SR) properties. In Section 3, we describe three classes of signed graphs obtained from a peculiar “alternating” join between the vertices of a signed graph and the corone of signed cycles. In Section 4, a similar construction is offered but instead of a vertex-wise join, it considers the join with a subgraph. In Section 5, we consider constructions based on the subdivision of a signed graph and related compound signed graphs. Finally, in Section 6, we derive the conclusions and offer an open problem.

1.1 Notations

The following notations will be used in the subsequent sections. The identity matrix of order n is denoted by I_n . The $m \times n$ matrix with all of its entries equal to one is denoted by $J_{m \times n}$ or just J_n if it is a square matrix with n rows and columns or simply J , where the relevant size should be easy enough to deduce. Similarly the $m \times n$ matrix with all of its entries equal to zero is denoted by $O_{m \times n}$ or just O_n if it is a square matrix with n rows and columns or simply O . We denote a column vector of size n with all its entries equal to one by $\mathbf{1}_n$. Likewise, a column vector of size n with all its entries equal to zero is denoted by $\mathbf{0}_n$. The column vector which has its i th entry equal to one, and all its other entries equal to zero are denoted by e_i . The matrix E_{ii} is defined as $e_i e_i^T$ and so will have its ii th entry equal to one and all others equal to zero. For any notations or results used further along in this article which have not been clearly stated, please refer to [10].

2 Basic results

A linear subgraph of G is a collection of vertex disjoint unions of cycles and edges of G . A matching of G is a collection of independent edges of G or in other words a set of edges of G without common vertices. A matching is a perfect matching if the matching edges span all the vertices of G .

Definition 2.1. Let p be any integer greater than or equal to 2 and let C_{2p} be the cycle graph of order $2p$. Let $\{u_1, u_2, \dots, u_p\}$ and $\{v_1, v_2, \dots, v_p\}$ be two disjoint vertex independent sets of the cycle C_{2p} . When considering these cycles as signed graphs we shall assign the value of +1 to every edge. Here, let us define the balanced corona cycle $H = C_{2p} \circ K_1$, where each newly introduced pendant edge is positive and let the pendant vertices in H adjacent to the vertices u_s and v_s , where $s = 1, 2, \dots, p$ be denoted by u'_s and v'_s , respectively.

Definition 2.2. Let q be any odd integer greater than or equal to 3 and let C_{2q} be the cycle graph of order $2q$. Let $\{\hat{u}_1, \hat{u}_2, \dots, \hat{u}_q\}$ and $\{\hat{v}_1, \hat{v}_2, \dots, \hat{v}_q\}$ be two disjoint vertex independent sets of the cycle C_{2q} . With proper vertex labelling as shown in Figure 1, let the q edges in the set $\{\hat{u}_x \hat{v}_y : x = y\}$ of the cycle C_{2q} have sign -1 and the remaining q edges have sign +1. Since q is an odd number the cycle shall be unbalanced. We will denote this signed and unbalanced cycle by $\hat{C}_{2q}$. Here, let us define the unbalanced corona cycle $\hat{H} = \hat{C}_{2q} \circ K_1$, where each

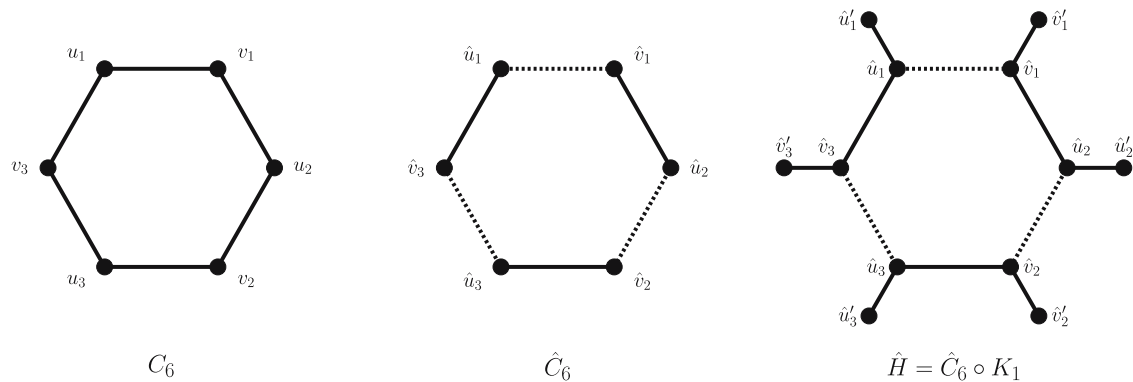


Figure 1: The graphs C_{2q} , $\hat{C}_{2q}$, and $\hat{H} = \hat{C}_{2q} \circ K_1$ for $q = 3$ with labelled vertices. Throughout this article positive edges will be represented by solid lines and negative edges will be represented by dashed lines.

newly introduced pendant edge is positive and let the pendant vertices in $\hat{H}$ adjacent to the vertices $\hat{u}_t$ or $\hat{v}_t$, where $t = 1, 2, \dots, q$ be denoted by $\hat{u}'_t$ and $\hat{v}'_t$ respectively.

Remark 2.3. The positive signs placed on the newly introduced pendant edges in Definitions 2.1 and 2.2 shall be important later on. The results obtained in the subsequent sections of this article generally do not hold for any arbitrary signature given to these edges.

Note 2.4. Note that the set of edges $\{\hat{u}_x \hat{v}_y : x = y\}$ and its complement $E(C_{2q}) \setminus \{\hat{u}_x \hat{v}_y : x = y\}$ form two equally sized distinct perfect matchings of the underlying cycle C_{2q} . As we can see in Figure 1 every vertex in an unbalanced cycle $\hat{C}_{2q}$ will be incident with precisely one positive edge and precisely one negative edge. Hence, the graph is net-regular of degree 0.

In [7], the authors consider the spectrum of the corona of two graphs. In particular, they show that for $G \circ K_1$, that is, the simple corona, the resulting graph does have the (-SR) property. The latter result is valid also in the context of signed graphs as well since the proof is not affected by the signs in G . We re-propose the result to keep the paper self-contained. We first need the following result, which can be found in several textbooks (see, for example, [4]).

Lemma 2.5. (Schur's complement formula) Let $A = \begin{pmatrix} P & Q \\ R & S \end{pmatrix}$ be a block matrix. Let P and S be square matrices. Then

- (a) If P is invertible, then $\det(A) = \det(P) \det(S - RP^{-1}Q)$.
- (b) If S is invertible, then $\det(A) = \det(S) \det(P - QS^{-1}R)$.

We can prove the following result on the simple corona of signed graphs.

Lemma 2.6. Let $\Sigma = (G, \sigma)$ be a signed graph and let $\Gamma = \Sigma \circ K_1$ be any simple corona, that is, the signed graph obtained by attaching a pendant vertex to each vertex of Σ with an edge of any sign. Then, Γ has the property (-SR).

Proof. Recall that any signing on bridges leads to switching equivalent signed graphs. Hence, it is not a restriction to consider only positive signs for the pendant edges of a simple corona graph, and this is the case we shall consider in the proof. With a suitable vertex ordering, the adjacency matrix of Γ has the following blockwise form:

$$A(\Gamma) = \begin{pmatrix} A(\Sigma) & I_n \\ I_n & O_n \end{pmatrix},$$

where I_n is the identity matrix and O_n is the null-matrix of order n .

Hence, by passing to the characteristic polynomial and by Schur's formula, we have:

$$\begin{aligned} P(\Gamma, x) &= \det(xI_{2n} - A(\Gamma)) \\ &= \det \begin{pmatrix} xI_n - A(\Sigma) & -I_n \\ -I_n & xI_n \end{pmatrix} \\ &= \det(xI_n) \det((xI_n - A(\Sigma)) - ((-I_n)(xI_n)^{-1}(-I_n))) \\ &= x^n \det \left(xI_n - A(\Sigma) - \frac{1}{x} I_n \right) \\ &= x^n \det \left(\left(x - \frac{1}{x} \right) I_n - A(\Sigma) \right) \\ &= x^n P \left(\Sigma, x - \frac{1}{x} \right). \end{aligned}$$

Hence, the characteristic polynomial of Γ satisfies property (-SR) if and only if $P \left(\Sigma, x - \frac{1}{x} \right)$ does as well.

Let λ be an eigenvalue of Σ . From $\lambda = x - \frac{1}{x}$, we obtain that $x = \frac{\lambda \pm \sqrt{\lambda^2 + 4}}{2}$. Therefore, for each eigenvalue λ of Σ , we obtain two eigenvalues of Γ such that their product is -1 , and Γ satisfies property (-SR). $\square$

The following corollary is a clear consequence of the previous lemma.

Corollary 2.7. *Let $\Sigma = (G, \sigma)$ be a signed graph with a symmetric spectrum and let $\Gamma = \Sigma \circ K_1$ be its simple corona. Then, Γ satisfies both properties (SR) and (-SR).*

In Figure 2 we show the simple corona $\Gamma = \Sigma \circ K_1$, where Σ has a symmetric spectrum. The signed graph Σ is taken from [11] as the smallest example of a signed graph with a symmetric spectrum which is not sign-symmetric (that is, Σ is not switching isomorphic to $-\Sigma$). We have

$$\text{Spec}(\Gamma) = \{\pm 2.600845_+, \pm 2.009693_+, \pm 1.543173_+, \pm 0.648015_+, \pm 0.497588_+, \pm 0.384490_+\}.$$

The spectrum of the signed unbalanced cycle, here denoted by $\hat{C}_n$, is as follows (see, for example, [2])

$$\text{Spec}(\hat{C}_n) = \left\{ 2 \cos \left(\frac{(2k+1)\pi}{n} \right) : k = 0, 1, \dots, n-1 \right\}.$$

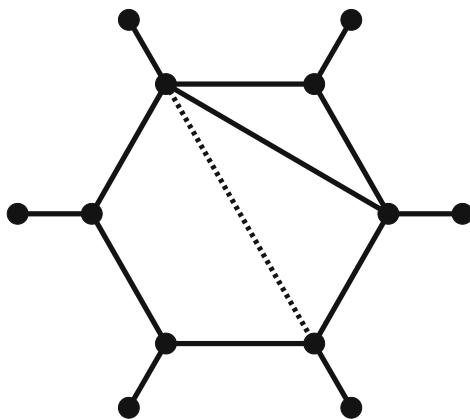


Figure 2: A signed graph with both properties (SR) and (-SR).

Note that the signed even cycles have a bipartite underlying graph, and so their spectrum is symmetric. Hence, we can deduce the following corollary.

Corollary 2.8. *The unbalanced corona cycle $\hat{H} = \hat{C}_{2q} \circ K_1$ satisfies both properties (SR) and (-SR).*

In [1], Ahmad et al. define polynomials of even degree which satisfy property (-SR).

Definition 2.9. [1] A polynomial $P(x) = \sum_{i=0}^{2n} a_i x^i$ is said to be an anti-reciprocal polynomial if it satisfies the following condition:

$$a_{2n-i} = \begin{cases} a_i & \text{if } i \text{ and } n \text{ have the same parity,} \\ -a_i & \text{otherwise,} \end{cases}$$

for all $i = 0, 1, 2, \dots, 2n$.

This is accompanied in [14] by Rakshith et al. with the following remark.

Remark 2.10. Let $P_1(x) = \sum_{i=0}^{2n} a_{2n-i} x^{2n-i}$ and $P_2(x) = \sum_{i=0}^{2n-2k} b_{2n-2k-i} x^{2n-2k-i}$ be two polynomials of degree $2n$ and $2n - 2k$, respectively. Let $P(x) = \sum_{i=0}^{2n} c_{2n-i} x^{2n-i} = P_1(x) + cx^k P_2(x)$, where c is any constant. Then

$$c_{2n-i} = \begin{cases} a_{2n-i} + cb_{2n-(i+k)} & \text{if } i = k, k+1, \dots, 2n-k, \\ a_{2n-i} & \text{otherwise.} \end{cases}$$

If both $P_1(x)$ and $P_2(x)$ satisfy the property (-SR), then

$$\begin{aligned} (-1)^n c_{2n-i} &= \begin{cases} (-1)^n a_{2n-i} + c(-1)^n b_{2n-(i+k)} & \text{if } i = k, k+1, \dots, 2n-k, \\ (-1)^n a_{2n-i} & \text{otherwise,} \end{cases} \\ &= \begin{cases} (-1)^i (a_i + cb_{i-k}) & \text{if } i = k, k+1, \dots, 2n-k, \\ (-1)^i a_i & \text{otherwise,} \end{cases} \\ &= (-1)^i c_i. \end{aligned}$$

Thus, $P(x)$ satisfies property (-SR).

The two following lemmas were obtained in [5] by Barik et al.

Lemma 2.11. [5] *Let G be an r -regular graph of order m and $H = G \circ K_1$. Then*

$$\mathbf{1}^T (xI_{2m} - A(H))^{-1} \mathbf{1} = \frac{(2x - r + 2)m}{x^2 - rx - 1}.$$

Lemma 2.12. [5] *Let A be a square matrix of order n , and let $A[1, 2, \dots, l]$ be the square matrix obtained by removing the 1st, 2nd, ..., l th rows and columns of the matrix A . Then for $1 \leq k \leq n$ and for constants $m_1, m_2, \dots, m_k$, we have*

$$\det \left(A + \sum_{i=1}^k m_i E_{ii} \right) = \det(A) + \sum_{i=1}^k \det(A[i]) + \sum_{\substack{i,j=1 \\ i < j}}^k m_i m_j \det(A[i, j]) + \dots + m_1 m_2 \dots m_k \det(A[1, 2, \dots, k]).$$

This next lemma is Lemma 3.7 in [14] with a slightly more general assumption. In the original variant, the matrix C is a $(0, 1)$ -matrix. For our purposes, we allow the matrix C in the subsequent lemma to be a $(-1, 0, 1)$ -square matrix. The proof is omitted as the original one remains valid under the given assumptions.

Lemma 2.13. Let C be a $(-1, 0, 1)$ -square matrix of order m such that $C\mathbf{1}_m = C^T\mathbf{1}_m = r\mathbf{1}_m$, where r is the row constant. Let Σ be a bipartite graph of order $2m$ with adjacency matrix $A(\Sigma) = \begin{pmatrix} O_m & C \\ C^T & O_m \end{pmatrix}$ and let $\Gamma = \Sigma \circ K_1$. If $E_m^T = (\mathbf{1}_m^T \ 0_m^T \ 0_m^T \ \mathbf{1}_m^T)$, then

$$E_m^T(xI_{4m} - A(\Gamma))^{-1}E_m = \frac{(2x^2 - r^2 + 2r - 2)mx}{x^4 - (r^2 + 2)x^2 + 1}.$$

Note here that in the case where $r = 0$, this becomes

$$E_m^T(xI_{4m} - A(\Gamma))^{-1}E_m = \frac{2mx}{x^2 - 1},$$

and in the case where $r = 2$, this becomes

$$E_m^T(xI_{4m} - A(\Gamma))^{-1}E_m = \frac{2mx(x^2 - 1)}{x^4 - 6x^2 + 1}.$$

3 Vertex-based joins

In this section, we analyse the first type of joins between a signed graph and the corone of cycles. The difference with [14] is that the root graph can be signed and the joining corone cycles are unbalanced. The three classes $(\mathcal{NC}_1^1)^-$, $(\mathcal{NC}_1^2)^-$, and $(\mathcal{NC}_1^3)^-$ represent three different ways of joining each vertex of the root graph with the vertices of the corone cycles.

3.1 Class $(\mathcal{NC}_1^1)^-$

Using the construction in Definition 2.2 consider the unbalanced corone cycle $\hat{H} = \hat{C}_{2q} \circ K_1$. Then the graph $\Gamma = G[\hat{C}_{2q}, n] \in (\mathcal{NC}_1^1)^-$ is obtained by taking n copies of $\hat{H} = \hat{C}_{2q} \circ K_1$ and joining the vertex $\hat{u}_j$ for $j = 1, 2, \dots, q$ in each copy of $\hat{C}_{2q} \circ K_1$ to a new vertex, say x , and by also joining all the pendant vertices adjacent to the vertices $\hat{v}_j$ for $j = 1, 2, \dots, q$ in each copy of $\hat{C}_{2q} \circ K_1$ to the vertex x and then joining this vertex x with another new vertex, say y .

In this class and those that follow, we have this pattern of constructing graphs where we have connecting edges coming from alternating vertices of a corone cycle – one in the cycle then one pendant and so on. For the benefit of tidy diagrams when this occurs, we will use a snaking line connected to a box surrounding the

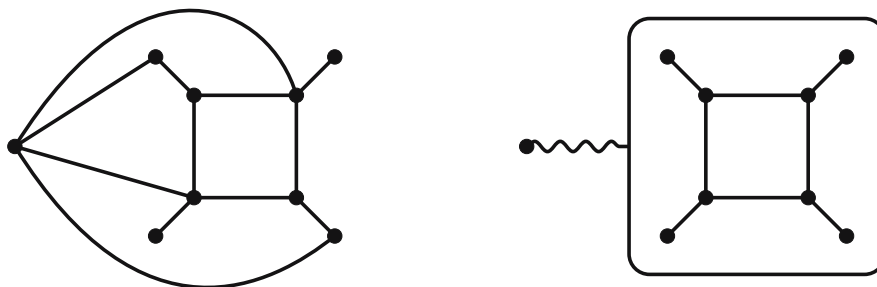


Figure 3: The graph on the right represents precisely the graph on the left. The snaking line connecting the isolated vertex to the box that surrounds the corone cycle represents edges between the isolated vertex and alternating cycle and pendant vertices of the corone cycle.

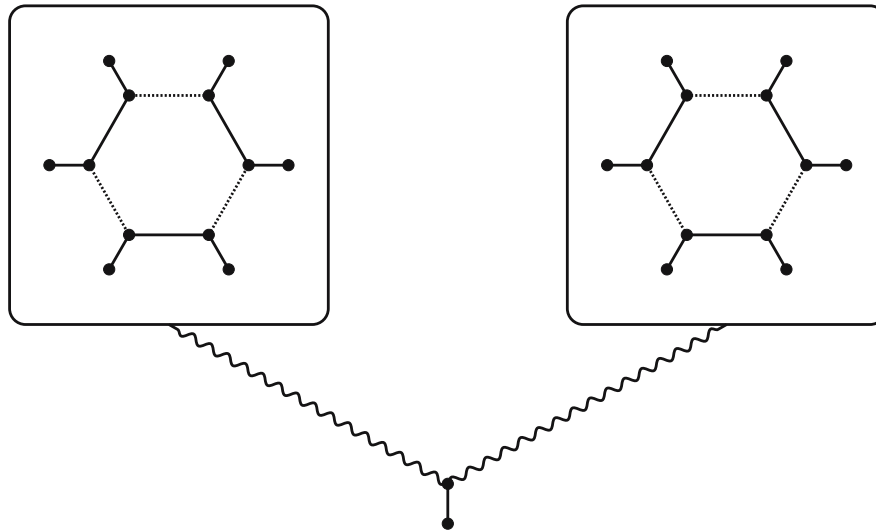


Figure 4: This is the graph $\Gamma_1 = G[\hat{C}_6, 2] \in (\mathcal{NC}_1^1)^-$.

corona cycle to represent edges to alternating vertices. Further along in the article, we will want to connect to all vertices of corona cycles, and in these instances, we will use a straight line in place of a snaking one. See Figure 3 for an example of this.

The following example illustrates that a specifically defined graph $\Gamma_1 \in (\mathcal{NC}_1^1)^-$ has the property (-SR).

Example 3.1. Let $\Gamma_1 = G[\hat{C}_6, 2] \in (\mathcal{NC}_1^1)^-$. This graph is shown in Figure 4. The spectrum of Γ_1 is made up of $+1, -1$ each with multiplicity 3; $\frac{\sqrt{7}}{2} + \frac{\sqrt{3}}{2}, \frac{\sqrt{7}}{2} - \frac{\sqrt{3}}{2}, -\frac{\sqrt{7}}{2} + \frac{\sqrt{3}}{2}, -\frac{\sqrt{7}}{2} - \frac{\sqrt{3}}{2}$ each with multiplicity 4; and $2 + \sqrt{3}, 2 - \sqrt{3}, -2 + \sqrt{3}, -2 - \sqrt{3}$ each with multiplicity 1. It is simple to check that this graph Γ_1 has the strong anti-reciprocal property (-SR).

Theorem 3.2. The graph $\Gamma \in (\mathcal{NC}_1^1)^-$ satisfies the property (-SR).

Proof. By proper vertex labelling of a graph $\Gamma \in (\mathcal{NC}_1^1)^-$, we have the adjacency matrix of Γ as follows:

$$A(\Gamma) = \begin{pmatrix} I_n \otimes \begin{pmatrix} O_{2q} & I_{2q} \\ I_{2q} & O_q & C \\ & C^T & O_q \end{pmatrix} & \mathbf{1}_n \otimes \begin{pmatrix} \mathbf{1}_q \\ \mathbf{0}_q \\ \mathbf{0}_q \\ \mathbf{1}_q \end{pmatrix} & \mathbf{0}_{4qn \times 1} \\ \mathbf{1}_n^T \otimes (\mathbf{1}_q^T \mathbf{0}_q^T \mathbf{0}_q^T \mathbf{1}_q^T) & 0 & 1 \\ \mathbf{0}_{1 \times 4qn} & 1 & 0 \end{pmatrix},$$

where $\begin{pmatrix} O_q & C \\ C^T & O_q \end{pmatrix}$ is the adjacency matrix of the signed cycle $\hat{C}_{2q}$, and we have -1 in each entry of the diagonal of the $q \times q$ matrix C .

Let v_i and w_i be the singular vector pairs corresponding to the singular values λ_i of C for $i = 1, 2, \dots, q$. Without loss of generality, we can assume that the sets $\{v_1, v_2, \dots, v_q\}$ and $\{w_1, w_2, \dots, w_q\}$ are orthonormal. Also since $C\mathbf{1}_q = \mathbf{0}_{1q}$ and $C^T\mathbf{1}_q = \mathbf{0}_{1q}$, we can further assume that $v_1 = w_1 = \frac{1}{\sqrt{q}}\mathbf{1}_q$.

Let $Z_i^+ = \begin{pmatrix} v_i^T & w_i^T & \delta_i v_i^T & \delta_i w_i^T \end{pmatrix}$ and $Z_i^- = \begin{pmatrix} v_i^T & -w_i^T & \delta'_i v_i^T & -\delta'_i w_i^T \end{pmatrix}$, where δ_i and δ'_i are some nonzero scalars. Then for $i = 2, 3, \dots, q$ and $j = 1, 2, \dots, n$, we obtain

$$A(\Gamma) \begin{pmatrix} e_j \otimes Z_i^+ \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} e_j \otimes \begin{pmatrix} v_i \\ w_i \\ \frac{1 + \lambda_i \delta_i}{\delta_i} v_i \\ \frac{1 + \lambda_i \delta_i}{\delta_i} w_i \end{pmatrix} \\ 0 \\ 0 \end{pmatrix} = \delta_i \begin{pmatrix} e_j \otimes \begin{pmatrix} v_i \\ w_i \\ \frac{1 + \lambda_i \delta_i}{\delta_i} v_i \\ \frac{1 + \lambda_i \delta_i}{\delta_i} w_i \end{pmatrix} \\ 0 \\ 0 \end{pmatrix}.$$

Therefore, $\begin{pmatrix} [e_j \otimes Z_i^+]^T, 0, 0 \end{pmatrix}$ is an eigenvector corresponding to an eigenvalue δ_i of $A(\Gamma)$ if and only if $\frac{1 + \lambda_i \delta_i}{\delta_i} = \delta_i$. That is, if and only if $\delta_i = \frac{\lambda_i \pm \sqrt{\lambda_i^2 + 4}}{2}$. Thus, $\frac{\lambda_i \pm \sqrt{\lambda_i^2 + 4}}{2}$ is an eigenvalue corresponding to the eigenvector $\begin{pmatrix} [e_j \otimes Z_i^+]^T, 0, 0 \end{pmatrix}$ of $A(\Gamma)$ with $\delta_i = \frac{\lambda_i \pm \sqrt{\lambda_i^2 + 4}}{2}$.

Similarly it can be seen that $\frac{-\lambda_i \pm \sqrt{\lambda_i^2 + 4}}{2}$ is an eigenvalue corresponding to the eigenvector $\begin{pmatrix} [e_j \otimes Z_i^-]^T, 0, 0 \end{pmatrix}$ of $A(\Gamma)$ with $\delta'_i = \frac{-\lambda_i \pm \sqrt{\lambda_i^2 + 4}}{2}$.

Now let $Z_1^+ = \begin{pmatrix} \mathbf{1}_q^T & \mathbf{1}_q^T & \delta_1 \mathbf{1}_q^T & \delta_1 \mathbf{1}_q^T \end{pmatrix}$ and $Z_1^- = \begin{pmatrix} \mathbf{1}_q^T & -\mathbf{1}_q^T & \delta'_1 \mathbf{1}_q^T & -\delta'_1 \mathbf{1}_q^T \end{pmatrix}$, where δ_1 and δ'_1 are some nonzero scalars.

Then for $j = 2, 3, \dots, n$, we have

$$A(\Gamma) \begin{pmatrix} (e_1 - e_j) \otimes Z_1^+ \\ 0 \\ 0 \end{pmatrix} = \delta_1 \begin{pmatrix} (e_1 - e_j) \otimes Z_1^+ \\ 0 \\ 0 \end{pmatrix} \Leftrightarrow \frac{1}{\delta_1} = \delta_1 \Leftrightarrow \delta_1 = \pm 1.$$

We also have

$$A(\Gamma) \begin{pmatrix} (e_1 - e_j) \otimes Z_1^- \\ 0 \\ 0 \end{pmatrix} = \delta'_1 \begin{pmatrix} (e_1 - e_j) \otimes Z_1^- \\ 0 \\ 0 \end{pmatrix} \Leftrightarrow \frac{1}{\delta'_1} = \delta'_1 \Leftrightarrow \delta'_1 = \pm 1.$$

Hence, $+1$ and -1 are eigenvalues of $A(\Gamma)$ with multiplicity $2(n-1)$ each.

Thus, we have listed $4qn - 4$ eigenvalues of $A(\Gamma)$ corresponding to the $4qn - 4$ orthogonal eigenvectors as described earlier.

Let t_i with $i = 1, 2, \dots, 6$ be the remaining eigenvalues of $A(\Gamma)$ corresponding to some eigenvectors X_i . Let $Y_i = \begin{pmatrix} [\mathbf{1}_n \otimes (e_4^i \otimes \mathbf{1}_q)]^T, 0, 0 \end{pmatrix}$, where $i = 1, 2, 3, 4$ and $Y_5 = \begin{pmatrix} \mathbf{0}_{4qn}^T, 1, 0 \end{pmatrix}$ and $Y_6 = \begin{pmatrix} \mathbf{0}_{4qn}^T, 0, 1 \end{pmatrix}$. Then, these Y_i with $i = 1, 2, \dots, 6$ along with the previously described $4qn - 4$ orthogonal eigenvectors of $A(\Gamma)$ form a set of $4qn + 2$ linearly independent vectors of size $4qn + 2$. Since X_i is an eigenvector corresponding to the eigenvalue t_i , then $X_i = \sum_{i=1}^6 a_i Y_i$ for some scalars a_i that are not all zero. Therefore, the equation $A(\Gamma)X_i = t_i X_i$ holds if and only if

$$\begin{aligned} a_1 t_i - a_3 - a_5 &= 0, \\ a_2 t_i - a_4 &= 0, \\ a_1 - a_3 t_i &= 0, \\ a_2 - a_4 t_i &= 0, \\ qna_1 + qna_4 - a_5 t_i + a_6 &= 0, \\ a_5 - a_6 t_i &= 0. \end{aligned}$$

Equivalently the equation $A(\Gamma)X_i = t_i X_i$ holds if and only if

$$\det \begin{pmatrix} -t_i & 0 & 1 & 0 & 1 & 0 \\ 0 & -t_i & 0 & 1 & 0 & 0 \\ 1 & 0 & -t_i & 0 & 0 & 0 \\ 0 & 1 & 0 & -t_i & 0 & 0 \\ qn & 0 & 0 & qn & -t_i & 1 \\ 0 & 0 & 0 & 0 & 1 & -t_i \end{pmatrix} = 0. \quad (1)$$

Thus, the remaining six eigenvalues of $A(\Gamma)$ are the roots of the polynomial

$$(t^2 - 1)(t^4 - 2(qn + 1)t^2 + 1).$$

Hence, the spectrum of the graph Γ consists of: $\frac{\lambda_i \pm \sqrt{\lambda_i^2 + 4}}{2}$ with multiplicity $n(q - 1)$; $\frac{-\lambda_i \pm \sqrt{\lambda_i^2 + 4}}{2}$ with multiplicity $n(q - 1)$; $+1$ with multiplicity $2(n - 1) + 1$; -1 with multiplicity $2(n - 1) + 1$; and the four roots of the polynomial $P(t) = t^4 - 2(qn + 1)t^2 + 1$. Since $P(t)$ satisfies the property (-SR) by Definition 2.9 and the remaining eigenvalues of Γ also satisfy the property (-SR), we can see that Γ has the anti-reciprocal property (-SR). $\square$

Remark 3.3. It appears that the property (-SR) is also satisfied for unbalanced cycles of length $2q$ when we take q to be any positive integer, not only the odd ones. However, the perfect matching method which gives the unbalanced cycle graph net-regularity of degree 0 invoked in our construction and used in this proof is not appropriate. This is because the cycles will then not actually be unbalanced as they will contain an even number of negative edges. Therefore, the proof for this is left open. This problem will again arise in the other classes further along in this article.

3.2 Class $(\mathcal{NC}_1^2)$

Let G_σ be any signed graph of order $N = n + m$ with vertex set $\{1, 2, \dots, n, n + 1, n + 2, \dots, n + m\}$ and then let $G_1 = G_\sigma \circ K_1$.

Consider the balanced corona cycles H_i from the construction in Definition 2.1 and the unbalanced corona cycles $\hat{H}_j$ from the construction in Definition 2.2.

The graph

$$\Gamma = G[G_1, H_1, H_2, \dots, H_n, \hat{H}_{n+1}, \hat{H}_{n+2}, \dots, \hat{H}_{n+m}] \in (\mathcal{NC}_1^2)^-$$

is obtained from the graphs G_1 , H_i ($i = 1, 2, \dots, n$), and $\hat{H}_j$ ($j = n + 1, n + 2, \dots, n + m$) by joining the vertex i ($i = 1, 2, \dots, n$) of G_σ with the vertices u_{is} and v'_{is} of H_i for $s = 1, 2, \dots, p_i$ and the vertex j ($j = n + 1, n + 2, \dots, n + m$) of G_σ with the vertices $\hat{u}_{jt}$ and $\hat{v}'_{jt}$ of $\hat{H}_j$ for $t = 1, 2, \dots, q_j$.

Example 3.4. Let $\Gamma_2 = G[G_1, H_1, H_2, \hat{H}_1] \in (\mathcal{NC}_1^2)^-$ with $G_1 = K_3 \circ K_1$, $H_1 = C_6 \circ K_1$, $H_2 = C_4 \circ K_1$, and $\hat{H}_1 = \hat{C}_6 \circ K_1$. This graph is shown in Figure 5. When factorised the characteristic polynomial of Γ_2 is

$$(x^2 - 1)^3(x^4 - 3x^2 + 1)^2(x^4 - 5x^2 + 1)^2(x^{16} - 35x^{14} - 2x^{13} + 418x^{12} + 26x^{11} - 1,989x^{10} - 100x^9 + 3,306x^8 + 100x^7 - 1,989x^6 - 26x^5 + 418x^4 + 2x^3 - 35x^2 + 1).$$

Hence, by Definition 2.9, the graph Γ_2 has the strong anti-reciprocal property (-SR).

Let $k = 4(p_1 + p_2 + \dots + p_n + q_{n+1} + q_{n+2} + \dots + q_{n+m})$. By proper vertex labelling of a graph $\Gamma \in (\mathcal{NC}_1^2)^-$, we have the adjacency matrix of Γ as follows:

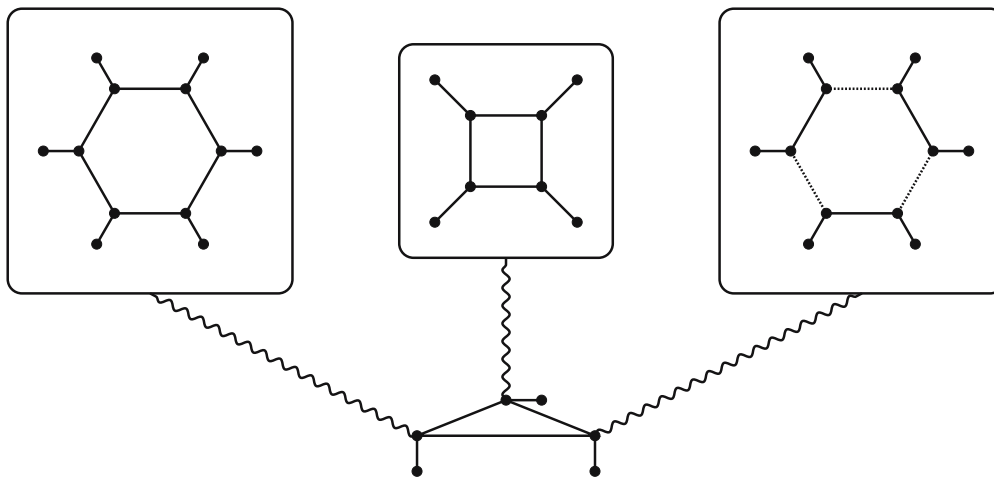


Figure 5: This is the graph $\Gamma_2 = G[G_1, H_1, H_2, \hat{H}_1] \in (\mathcal{NC}_1^2)^-$ with $G_1 = K_3 \circ K_1$, $H_1 = C_6 \circ K_1$, $H_2 = C_4 \circ K_1$, and $\hat{H}_1 = \hat{C}_6 \circ K_1$.

$$A(\Gamma) = \begin{pmatrix} B & E & O_{k \times N} \\ E^T & A(G_\sigma) & I_N \\ O_{N \times k} & I_N & O_N \end{pmatrix},$$

where B is the square matrix of order k and E is the matrix of order $k \times N$ as defined below

$$B = \begin{pmatrix} A(H_1) & O & \cdots & O & O & O & \cdots & O \\ O & A(H_2) & \cdots & O & O & O & \cdots & O \\ \vdots & \vdots & \ddots & O & O & O & \cdots & O \\ O & O & O & A(H_n) & O & O & \cdots & O \\ O & O & O & O & A(\hat{H}_{n+1}) & O & \cdots & O \\ O & O & O & O & O & A(\hat{H}_{n+2}) & \cdots & O \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & O \\ O & O & O & O & O & O & O & A(\hat{H}_{n+m}) \end{pmatrix},$$

$$E = \begin{pmatrix} E_{p_1} & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ 0 & E_{p_2} & \cdots & 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & E_{p_n} & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 0 & E_{q_{n+1}} & 0 & \cdots & 0 \\ 0 & 0 & 0 & 0 & 0 & E_{q_{n+2}} & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & E_{q_{n+m}} \end{pmatrix} \quad \text{with } E_{p_i} = \begin{pmatrix} 1_{p_i} \\ 0_{p_i} \\ 0_{p_i} \\ 1_{p_i} \end{pmatrix} \quad \text{and} \quad E_{q_j} = \begin{pmatrix} 1_{q_j} \\ 0_{q_j} \\ 0_{q_j} \\ 1_{q_j} \end{pmatrix}.$$

Theorem 3.5. The graph $\Gamma \in (\mathcal{NC}_1^2)^-$ satisfies the property (-SR).

Proof. From the adjacency matrix of the graph Γ , we obtain

$$\det(xI - A(\Gamma)) = \det \begin{pmatrix} xI_k - B & -E & O_{k \times N} \\ -E^T & xI_N - A(G_\sigma) & -I_N \\ O_{N \times k} & -I_N & xI_N \end{pmatrix}.$$

Applying Schur's formula as in Lemma 2.5 to this determinant, we obtain

$$\begin{aligned}
P(\Gamma; x) &= \det(xI_k - B) \det \left(\begin{pmatrix} xI_N - A(G_\sigma) & -I_N \\ -I_N & xI_N \end{pmatrix} - \begin{pmatrix} -E^T \\ O_{N \times k} \end{pmatrix} (xI_k - B)^{-1} \begin{pmatrix} -E^T \\ O_{N \times k} \end{pmatrix}^T \right) \\
&= \prod_{i=1}^n P(H_i; x) \prod_{j=n+1}^{n+m} P(\hat{H}_j; x) \times \det \left(\begin{pmatrix} xI_N - A(G_\sigma) & -I_N \\ -I_N & xI_N \end{pmatrix} - \begin{pmatrix} X & O_N \\ O_N & O_N \end{pmatrix} \right),
\end{aligned}$$

where

$$X = \sum_{i=1}^n (E_{p_i}^T (xI_{4p_i} - A(H_i))^{-1} E_{p_i}) E_{ii} + \sum_{j=n+1}^{n+m} (E_{q_j}^T (xI_{4q_j} - A(\hat{H}_j))^{-1} E_{q_j}) E_{jj}.$$

By Lemma 2.13,

$$X = \sum_{i=1}^n \frac{2p_i x(x^2 - 1)}{x^4 - 6x^2 + 1} E_{ii} + \sum_{j=n+1}^{n+m} \frac{2q_j x}{x^2 - 1} E_{jj}.$$

By combining this with Lemma 2.12, we obtain

$$\begin{aligned}
P(\Gamma; x) &= \prod_{i=1}^n P(H_i; x) \prod_{j=n+1}^{n+m} P(\hat{H}_j; x) \times \left[\det(xI_{2N} - A(G_1)) + \frac{2x(1 - x^2)}{x^4 - 6x^2 + 1} \sum_{i=1}^n p_i \det(xI_{2N} - A(G_1)[i]) \right. \\
&\quad - \frac{2x}{x^2 - 1} \sum_{j=n+1}^{n+m} q_j \det(xI_{2N} - A(G_1)[j]) + \left(\frac{2x(1 - x^2)}{x^4 - 6x^2 + 1} \right)^2 \sum_{\substack{i, \tilde{i}=1 \\ i < \tilde{i}}}^n p_i p_{\tilde{i}} \det(xI_{2N} - A(G_1)[i, \tilde{i}]) \\
&\quad - \left(\frac{2x}{x^2 - 1} \right)^2 \sum_{\substack{j, \tilde{j}=n+1 \\ j < \tilde{j}}}^{n+m} q_j q_{\tilde{j}} \det(xI_{2N} - A(G_1)[j, \tilde{j}]) + \dots \\
&\quad + \left(\frac{2x(1 - x^2)}{x^4 - 6x^2 + 1} \right)^n p_1 p_2 \dots p_n \det(xI_{2N} - A(G_1)[1, 2, \dots, n]) \\
&\quad \left. - \left(\frac{2x}{x^2 - 1} \right)^m q_{n+1} q_{n+2} \dots q_{n+m} \det(xI_{2N} - A(G_1)[n+1, n+2, \dots, n+m]) \right].
\end{aligned}$$

Therefore,

$$\begin{aligned}
P(\Gamma; x) &= \prod_{i=1}^n P(H_i; x) \prod_{j=n+1}^{n+m} P(\hat{H}_j; x) \times \left[(x^4 - 6x^2 + 1)^n P(G_1; x) + (x^2 - 1)^m P(G_1; x) + 2x(1 - x^2)(x^4 - 6x^2 + 1)^{n-1} \right. \\
&\quad \times \sum_{i=1}^n p_i P(G_1 \setminus \{i, i'\}; x) - 2x(x^2 - 1)^{m-1} \sum_{j=n+1}^{n+m} q_j P(G_1 \setminus \{j, j'\}; x) \\
&\quad + 4x^2(1 - x^2)^2(x^4 - 6x^2 + 1)^{n-2} \sum_{\substack{i, \tilde{i}=1 \\ i < \tilde{i}}}^n p_i p_{\tilde{i}} P(G_1 \setminus \{i, i', \tilde{i}, \tilde{i}'\}; x) \\
&\quad \left. - 4x^2(x^2 - 1)^{m-2} \sum_{\substack{j, \tilde{j}=n+1 \\ j < \tilde{j}}}^{n+m} q_j q_{\tilde{j}} P(G_1 \setminus \{j, j', \tilde{j}, \tilde{j}'\}; x) + \dots + 2^n x^n (1 - x^2)^n p_1 p_2 \dots p_n - 2^m x^m q_{n+1} q_{n+2} \dots q_{n+m} \right].
\end{aligned}$$

Note that for $\tilde{i} = 1, 2, \dots, n-1$ and $i_a \in \{1, 2, \dots, n\}$, we obtain that

$$G_1 \setminus \{i_1, i_2, \dots, i_{\tilde{i}}, i'_1, i'_2, \dots, i'_{\tilde{i}}\}$$

is a corona graph, and hence, the polynomial $P(G_1 \setminus \{i_1, i_2, \dots, i_{\tilde{i}}, i'_1, i'_2, \dots, i'_{\tilde{i}}\}; x)$ satisfies the property (-SR).

Likewise for $\tilde{j} = n+1, n+2, \dots, n+m-1$ and $j_b \in \{n+1, n+2, \dots, n+m\}$, we obtain that

$$G_1 \setminus \{j_{n+1}, j_{n+2}, \dots, j_{n+\bar{j}}, j'_{n+1}, j'_{n+2}, \dots, j'_{n+\bar{j}}\}$$

is a corona graph, and hence, the polynomial $P(G_1 \setminus \{j_{n+1}, j_{n+2}, \dots, j_{n+\bar{j}}, j'_{n+1}, j'_{n+2}, \dots, j'_{n+\bar{j}}\}; x)$ satisfies the property (-SR).

Since

$$\frac{P(H_i; x)}{x^4 - 6x^2 + 1} \quad \text{and} \quad (1 - x^2)^i (x^4 - 6x^2 + 1)^{n-i} P(G_1 \setminus \{i_1, i_2, \dots, i_{\bar{i}}, i'_1, i'_2, \dots, i'_{\bar{i}}\}; x),$$

$$\frac{P(\hat{H}_j; x)}{x^2 - 1} \quad \text{and} \quad (x^2 - 1)^{m-j} P(G_1 \setminus \{j_{n+1}, j_{n+2}, \dots, j_{n+\bar{j}}, j'_{n+1}, j'_{n+2}, \dots, j'_{n+\bar{j}}\}; x),$$

all satisfy the property (-SR) the theorem follows from Remark 2.10. $\square$

3.3 Class $(\mathcal{NC}_1^3)^-$

Let $F_k = C_{2r_k} \circ K_1$ be a balanced corona cycle constructed in the same style as $H_i = C_{2p_i} \circ K_1$ according to Definition 2.1.

Let G_σ be any signed graph of order $N = n + m + l$ with vertex set $\{1, 2, \dots, n, n+1, n+2, \dots, n+m, n+m+1, n+m+2, \dots, n+m+l\}$ and then let $G_1 = G_\sigma \circ K_1$. The graph

$$\Gamma = G[G_1, H_1, H_2, \dots, H_n, \hat{H}_{n+1}, \hat{H}_{n+2}, \dots, \hat{H}_{n+m}, F_{n+m+1}, F_{n+m+2}, \dots, F_{n+m+l}] \in (\mathcal{NC}_1^3)^-$$

is obtained from the graphs G_1 , H_a ($a = 1, 2, \dots, n$), $\hat{H}_b$ ($b = n+1, n+2, \dots, n+m$), and F_c ($c = n+m+1, n+m+2, \dots, n+m+l$) by joining the vertex a ($a = 1, 2, \dots, n$) of G_σ with the vertices u_{as} and v'_{as} of H_a for $s = 1, 2, \dots, p_a$ and the vertex b ($b = n+1, n+2, \dots, n+m$) of G_σ with the vertices $\hat{u}_{bt}$ and $\hat{v}'_{bt}$ of $\hat{H}_b$ for $t = 1, 2, \dots, q_b$ and the vertex c ($c = n+m+1, n+m+2, \dots, n+m+l$) of G_σ with all vertices – both those vertices in the cycles and those that are pendants – of each and every F_c .

It should be noted that this class $(\mathcal{NC}_1^3)^-$ is a generalisation of the previous class $(\mathcal{NC}_1^2)^-$. Following a similar method as in the proof of Theorem 3.5, we obtain the following theorem and thus a third class of graphs satisfying the property (-SR).

Theorem 3.6. *The graph $\Gamma \in (\mathcal{NC}_1^3)^-$ satisfies the property (-SR).*

4 Subgraph based joins

In this section, we consider the joins between the non-pendant vertices of the corona of a net-regular signed graph with the corone of balanced and unbalanced cycles. Following [14], we have the two classes $(\mathcal{NC}_2^1)^-$ and $(\mathcal{NC}_2^2)^-$.

4.1 Class $(\mathcal{NC}_2^1)^-$

Let $H_a = C_{2p_a} \circ K_1$ and $F_c = C_{2l_c} \circ K_1$ be balanced corona cycles and let $\hat{H}_b = \hat{C}_{2q_b} \circ K_1$ be unbalanced corona cycles in the way with which we are now familiar.

For any signed graph G_σ that is net-regular of degree r and is of order N let $G_1 = G_\sigma \circ K_1$. The graph

$$\Gamma = G[G_1, H_1, H_2, \dots, H_n, \hat{H}_1, \hat{H}_2, \dots, \hat{H}_m, F_1, F_2, \dots, F_k] \in (\mathcal{NC}_2^1)^-$$

is obtained from the graphs G_1 , H_a ($a = 1, 2, \dots, n$), $\hat{H}_b$ ($b = 1, 2, \dots, m$), and F_c ($c = 1, 2, \dots, k$) by joining all the vertices of G_σ with the vertices u_{as} and v'_{as} of each H_a for $s = 1, 2, \dots, p_a$ and with the vertices $\hat{u}_{bt}$ and $\hat{v}'_{bt}$ of each $\hat{H}_b$ for $t = 1, 2, \dots, q_b$ and with all vertices – both those vertices in the cycles and those that are pendants – of each and every F_c . Note that the order N of the graph G_σ is not reliant on n and m and k here.

Example 4.1. Let $\Gamma_3 = G[G_1, H_1, \hat{H}_1, F_1] \in (\mathcal{NC}_2^1)^-$ with $G_1 = K_3 \circ K_1$, $H_1 = C_4 \circ K_1$, $\hat{H}_1 = \hat{C}_6 \circ K_1$, and $F_1 = C_4 \circ K_1$. This graph is shown in Figure 6. When factorised, the characteristic polynomial of Γ_3 is

$$(x^2 - 1)^5(x^2 + x - 1)(x^4 - 5x^2 + 1)(x^4 - 6x^2 + 1)(x^6 + x^5 - 6x^4 - 5x^3 + 6x^2 + 1 - 1) \\ (x^8 - 2x^7 - 62x^6 - 34x^5 + 194x^4 + 34x^3 - 62x^2 + 2x + 1).$$

Hence, by Definition 2.9, the graph Γ_3 has the strong anti-reciprocal property (-SR).

Let $P = 4(p_1 + p_2 + \dots + p_n)$ and $Q = 4(q_1 + q_2 + \dots + q_m)$ and $L = 4(l_1 + l_2 + \dots + l_k)$. Then let $k_1 = P + Q$ and $k_2 = L$. By proper vertex labelling of a graph $\Gamma \in (\mathcal{NC}_2^1)^-$, its adjacency matrix can be written as follows:

$$A(\Gamma) = \begin{pmatrix} H & O_{k_1 \times k_2} & E & O_{k_1 \times N} \\ O_{k_2 \times k_1} & F & J_{k_2 \times N} & O_{k_2 \times N} \\ E^T & J_{N \times k_2} & A(G_\sigma) & I_N \\ O_{N \times k_1} & O_{N \times k_2} & I_N & O_N \end{pmatrix},$$

where J is the all one matrix, H is the square matrix of order k_1 , and F is the square matrix of order k_2 , and E is the matrix of order $k_1 \times N$ as defined below

$$H = \begin{pmatrix} A(H_1) & O & \cdots & O & O & O & \cdots & O \\ O & A(H_2) & \cdots & O & O & O & \cdots & O \\ \vdots & \vdots & \ddots & O & O & O & \cdots & O \\ O & O & O & A(H_n) & O & O & \cdots & O \\ O & O & O & O & A(\hat{H}_1) & O & \cdots & O \\ O & O & O & O & O & A(\hat{H}_2) & \cdots & O \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & O \\ O & O & O & O & O & O & O & A(\hat{H}_m) \end{pmatrix},$$

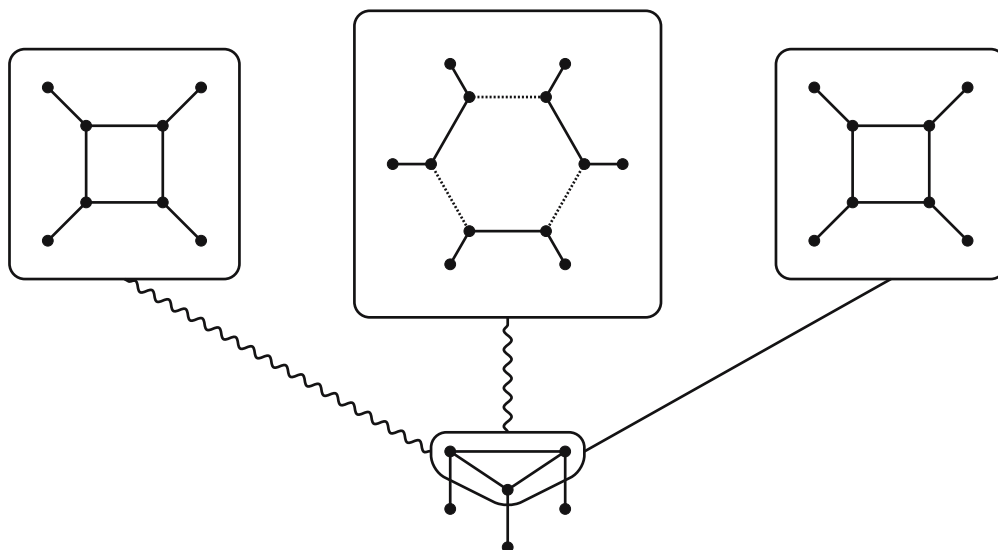


Figure 6: This is the graph $\Gamma_3 = G[G_1, H_1, \hat{H}_1, F_1] \in (\mathcal{NC}_2^1)^-$ with $G_1 = K_3 \circ K_1$, $H_1 = C_4 \circ K_1$, $\hat{H}_1 = \hat{C}_6 \circ K_1$, and $F_1 = C_4 \circ K_1$. The edges between G_1 and H_1 , $\hat{H}_1$, F_1 will each connect to all three of the vertices in K_3 .

$$F = \begin{pmatrix} A(F_1) & O & \cdots & O & O \\ O & A(F_2) & \cdots & O & O \\ \vdots & \vdots & \ddots & O & O \\ O & O & O & A(F_{k-1}) & O \\ O & O & O & O & A(F_k) \end{pmatrix},$$

$$E = \begin{pmatrix} E_{p_1} & E_{p_1} & \cdots & E_{p_1} & E_{p_1} \\ E_{p_2} & E_{p_2} & \cdots & E_{p_2} & E_{p_2} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ E_{p_n} & E_{p_n} & \cdots & E_{p_n} & E_{p_n} \\ E_{q_1} & E_{q_1} & \cdots & E_{q_1} & E_{q_1} \\ E_{q_2} & E_{q_2} & \cdots & E_{q_2} & E_{q_2} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ E_{q_m} & E_{q_m} & \cdots & E_{q_m} & E_{q_m} \end{pmatrix} \quad \text{with} \quad E_{p_a} = \begin{pmatrix} \mathbf{1}_{p_a} \\ \mathbf{0}_{p_a} \\ \mathbf{0}_{p_a} \\ \mathbf{1}_{p_a} \end{pmatrix} \quad \text{and} \quad E_{q_b} = \begin{pmatrix} \mathbf{1}_{q_b} \\ \mathbf{0}_{q_b} \\ \mathbf{0}_{q_b} \\ \mathbf{1}_{q_b} \end{pmatrix}.$$

Theorem 4.2. The graph $\Gamma \in (\mathcal{NC}_2^1)^-$ satisfies the property (-SR).

Proof. From the adjacency matrix of the graph Γ , we obtain

$$\det(xI - A(\Gamma)) = \det \begin{pmatrix} xI_{k_1} - H & O_{k_1 \times k_2} & -E & O_{k_1 \times N} \\ O_{k_2 \times k_1} & xI_{k_2} - F & -J_{k_2 \times N} & O_{k_2 \times N} \\ -E^T & -J_{N \times k_2} & xI_N - A(G_\sigma) & -I_N \\ O_{N \times k_1} & O_{N \times k_2} & -I_N & xI_N \end{pmatrix}.$$

By applying Schur's formula as in Lemma 2.5 to this determinant, we obtain

$$\begin{aligned} \det(xI - A(\Gamma)) &= \det \begin{pmatrix} xI_{k_1} - H & O_{k_1 \times k_2} \\ O_{k_2 \times k_1} & xI_{k_2} - F \end{pmatrix} \times \det \begin{pmatrix} xI_N - A(G_\sigma) & -I_N \\ -I_N & xI_N \end{pmatrix} \\ &\quad - \begin{pmatrix} -E^T & -J_{N \times k_2} \\ O_{N \times k_1} & O_{N \times k_2} \end{pmatrix} \begin{pmatrix} xI_{k_1} - H & O_{k_1 \times k_2} \\ O_{k_2 \times k_1} & xI_{k_2} - F \end{pmatrix}^{-1} \begin{pmatrix} -E & O_{k_1 \times N} \\ -J_{k_2 \times N} & O_{k_2 \times N} \end{pmatrix} \\ &= \det((xI_{k_1} - H)(xI_{k_2} - F)) \times \det \begin{pmatrix} xI_N - A(G_\sigma) & -I_N \\ -I_N & xI_N \end{pmatrix} \\ &\quad - \begin{pmatrix} E^T(xI_{k_1} - H)^{-1}E + J_{N \times k_2}(xI_{k_2} - F)^{-1}J_{k_2 \times N} & O_N \\ O_N & O_N \end{pmatrix} \\ &= \det((xI_{k_1} - H)(xI_{k_2} - F)) \times \det \begin{pmatrix} xI_N - A(G_\sigma) - (\alpha + \beta)J_N & -I_N \\ -I_N & xI_N \end{pmatrix}, \end{aligned}$$

where

$$\alpha = E^T(xI_{k_1} - H)^{-1}E = \sum_{a=1}^n E_{p_a}^T(xI_{4p_a} - A(H_a))^{-1}E_{p_a} + \sum_{b=1}^m E_{q_b}^T(xI_{4q_b} - A(\hat{H}_b))^{-1}E_{q_b},$$

$$\beta = J_{N \times k_2}(xI_{k_2} - F)^{-1}J_{k_2 \times N} = \sum_{c=1}^k \mathbf{1}_{4l_c}^T(xI_{4l_c} - A(F_c))^{-1}\mathbf{1}_{4l_c}.$$

Now by Lemmas 2.11 and 2.13, we obtain

$$\alpha = \frac{Px(x^2 - 1)}{2(x^4 - 6x^2 + 1)} + \frac{Qx}{2(x^2 - 1)} \quad \text{and} \quad \beta = \frac{Lx}{x^2 - 2x - 1}.$$

Reapplying Schur's formula gives us that $\det(xI - A(\Gamma))$

$$\begin{aligned} &= \det((xI_{k_1} - H)(xI_{k_2} - F)) \det(xI_N) \det((xI_N - A(G_\sigma) - (\alpha + \beta)J_N) - (-I_N)(xI_N)^{-1}(-I_N)) \\ &= \det((xI_{k_1} - H)(xI_{k_2} - F)) \times x^N \times \det\left(\left(x - \frac{1}{x}\right)I_N - A(G_\sigma) - (\alpha + \beta)J_N\right). \end{aligned}$$

Since G_σ is an r -regular graph, we can see that the matrices $\left(x - \frac{1}{x}\right)I_N - (A(G_\sigma) + (\alpha + \beta)J_N)$ and $\left(x - \frac{1}{x}\right)I_N - D$ are similar where D is the diagonal matrix given by $D = \text{diag}(r + N(\alpha + \beta), \lambda_2, \lambda_3, \dots, \lambda_N)$. It will also be useful to note that $x^N = x \times \prod_{i=2}^N x$. Thus, the equation for $\det(xI - A(\Gamma))$ reduces to

$$\begin{aligned} \det(xI - A(\Gamma)) &= \det((xI_{k_1} - H)(xI_{k_2} - F)) \times x^N \times \det\left(\left(x - \frac{1}{x}\right)I_N - D\right) \\ &= \det((xI_{k_1} - H)(xI_{k_2} - F)) \times x^N \times \left(x - \frac{1}{x} - r - N(\alpha + \beta)\right) \\ &\quad \left(x - \frac{1}{x} - \lambda_2\right) \left(x - \frac{1}{x} - \lambda_3\right) \cdots \left(x - \frac{1}{x} - \lambda_N\right) \\ &= \det((xI_{k_1} - H)(xI_{k_2} - F)) \times x \times \left(x - \frac{1}{x} - r - N(\alpha + \beta)\right) \prod_{i=2}^N x \times \prod_{i=2}^N \left(x - \frac{1}{x} - \lambda_i\right) \\ &= \det((xI_{k_1} - H)(xI_{k_2} - F)) R(x) \prod_{i=2}^N (x^2 - \lambda_i x - 1), \end{aligned}$$

where

$$\begin{aligned} R(x) &= x^2 - 1 - rx - Nx(\alpha + \beta) \\ &= x^2 - 1 - rx - Nx \left(\frac{Px(x^2 - 1)}{2(x^4 - 6x^2 + 1)} + \frac{Qx}{2(x^2 - 1)} + \frac{Lx}{x^2 - 2x - 1} \right) \\ &= (2x^6 - 14x^4 + 14x^2 - 2)^{-1} \times [2x^8 - 2rx^7 - (PN + QN + 2LN + 16)x^6 \\ &\quad - (4LN - 14r)x^5 + (2PN + 6QN + 4LN + 28)x^4 + (4LN - 14r)x^3 \\ &\quad - (PN + QN + 2LN + 16)x^2 + 2rx + 2]. \end{aligned}$$

Therefore,

$$P(\Gamma; x) = R(x) \prod_{a=1}^n P(H_a; x) \prod_{b=1}^m P(\hat{H}_b; x) \prod_{c=1}^k P(F_c; x) \prod_{i=2}^N (x^2 - \lambda_i x - 1).$$

Since H_a and $\hat{H}_b$ and F_c are corona graphs, their characteristic polynomials will satisfy the property (-SR). By Definition 2.9, we know that $R(x)$ satisfies the property (-SR). Putting all of these elements together, we see that the graph $\Gamma \in (\mathcal{NC}_2^1)^-$ satisfies the property (-SR). $\square$

4.2 Class $(\mathcal{NC}_2^2)$

The join of two ordinary graphs G and H is denoted by $G \vee H$ and is the graph obtained by joining each vertex of G with every vertex of H . For two signed graphs G_σ and H_τ , the positive join is denoted by $G_\sigma \vee_+ H_\tau$ and is the signed graph obtained by connecting each vertex of G_σ to every vertex of H_τ with a positive edge.

Let G_σ and K_τ be two net-regular signed graphs such that their positive join $G_\sigma \vee_+ K_\tau$ is also net-regular. They are not required to be net-regular of the same degree, and in fact, it is not guaranteed that any two net-regular signed graphs will have a net-regular positive join. Let $G_1 = (G_\sigma \vee_+ K_\tau) \circ K_1$. The graph

$$\Gamma = G[G_1, H_1, H_2, \dots, H_n, \hat{H}_1, \hat{H}_2, \dots, \hat{H}_m, F_1, F_2, \dots, F_k] \in (\mathcal{NC}_2^-)$$

is obtained from the graphs G_1 , H_a ($a = 1, 2, \dots, n$), $\hat{H}_b$ ($b = 1, 2, \dots, m$), and F_c ($c = 1, 2, \dots, k$) by joining all the vertices of G_σ with the vertices u_{as} and v'_{as} of each H_a for $s = 1, 2, \dots, p_a$ and with the vertices $\hat{u}_{bt}$ and $\hat{v}'_{bt}$ of each $\hat{H}_b$ for $t = 1, 2, \dots, q_b$ and by joining all the vertices of K_τ with all vertices – both those vertices in the cycles and those that are pendants – of each and every F_c .

The following theorem can be obtained following a similar strategy to that employed in the proof of Theorem 4.2.

Theorem 4.3. *The graph $\Gamma \in (\mathcal{NC}_2^-)$ satisfies the property (-SR).*

Remark 4.4. It appears that if we change the construction of a graph $\Gamma \in (\mathcal{NC}_2^-)$ so that G_σ can be any signed graph and not just a net-regular signed graph then the property (-SR) is still satisfied. The same applies to the signed graphs G_σ and K_τ in the construction of a graph $\Gamma \in (\mathcal{NC}_2^-)$. That is, the anti-reciprocal property is retained when G_σ and K_τ and also $G_\sigma \vee K_\tau$ are not net-regular. However, the details in the proofs of the theorems in this section do rely on the net-regularity of these graphs. Hence, this problem is left open.

5 Subdivision-like based joins

The final classes of noncorona signed graphs with strong anti-reciprocal eigenvalue property to be presented in this article make use of several other graphs which are obtained from one root graph. In fact, from a graph G , we can derive the subdivision graph $S(G)$, the total graph $T(G)$, and two other intermediate graphs denoted in [10] as $R(G)$ and $Q(G)$. We first introduce the compound graphs for unsigned graphs and then we define the signed variants. Also in this case regularity is required in order to achieve the whole spectrum.

5.1 Ordinary graphs

For an undirected simple graph G with vertex set V of order n and edge set E of size m , the vertex-edge incidence matrix of G is the $n \times m$ matrix denoted B such that $B_{ij} = 1$ if the vertex v_i is incident with the edge e_j and 0 otherwise.

The line graph $L(G)$ of an undirected simple graph G is the graph whose vertex set is in one-to-one correspondence with the set of edges of the graph G and which has two vertices being adjacent if and only if the corresponding edges in G have a vertex in common.

The subdivision graph $S(G)$ of a graph G is the graph obtained by inserting a new vertex into every edge of G . Equivalently we replace each edge of G with a path of length 2. Recall from say [10] that the subdivision graph is a bipartite graph whose adjacency matrix is of the form

$$A_{S(G)} = \begin{pmatrix} O & B \\ B^T & O \end{pmatrix}.$$

Let $Q(G)$ be the graph obtained from $S(G)$ by adding an edge between two of the newly inserted vertices whenever the corresponding edges are incident in G . The adjacency matrix of $Q(G)$ is then of the form

$$A_{Q(G)} = \begin{pmatrix} O & B \\ B^T & A_{L(G)} \end{pmatrix}.$$

Let $R(G)$ be the graph obtained from G by introducing a new vertex corresponding to each edge of G and by joining each new vertex to the end vertices of the edge it corresponds to. Equivalently we replicate each edge of G by adding another path of length 2 between each pair of vertices adjacent in G . The adjacency matrix of $R(G)$ is of the form

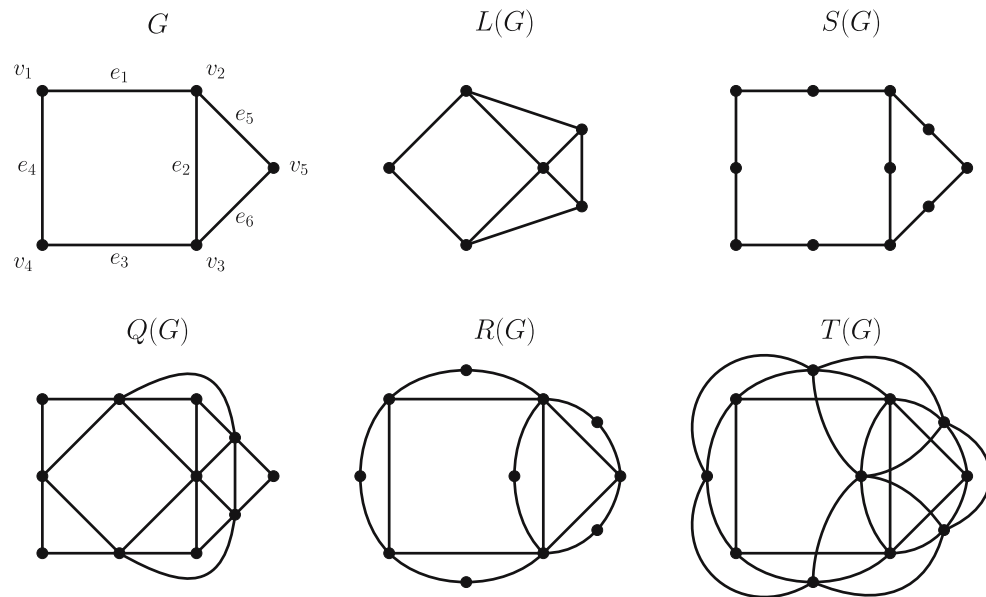


Figure 7: A graph G along with its line graph $L(G)$, its subdivision graph $S(G)$, the related graphs $Q(G)$ and $R(G)$, and its total graph $T(G)$.

$$A_{R(G)} = \begin{pmatrix} A_G & B \\ B^T & O \end{pmatrix}.$$

The total graph $T(G)$ is obtained from $R(G)$ by adding an edge between two of the newly introduced vertices whenever the corresponding edges are incident in G . In other words, it is the graph obtained from G by combining the adjacency matrix of the graph G with the adjacency matrix of its line graph and its vertex-edge incidence matrix. Precisely the adjacency matrix of $T(G)$ is of the form

$$A_{T(G)} = \begin{pmatrix} A_G & B \\ B^T & A_{L(G)} \end{pmatrix}.$$

For examples of these graphs see Figure 7.

5.2 Signed graphs

In [16], Zaslavsky gave the first definition of the incidence matrix for signed graphs. As further discussed by Belardo et al. in [9], this is a necessary step in obtaining a spectrally consistent definition of a signed line graph.

For a signed graph $\Sigma = G_\sigma$, we introduce the vertex-edge orientation $\eta : V(G) \times E(G) \rightarrow \{-1, 0, 1\}$ formed by obeying the following rules:

- (O1) $\eta(i, jk) = 0$ if $i \notin \{j, k\}$;
- (O2) $\eta(i, ij) = 1$ or $\eta(i, ij) = -1$;
- (O3) $\eta(i, ij)\eta(j, ij) = -\sigma(ij)$.

The incidence matrix $B_\eta = (\eta_{ij})$ is a vertex-edge incidence matrix derived from G_σ such that its (i, e) -entry is equal to $\eta(i, e)$. However, it is not uniquely determined by G_σ alone. As in the definition of the oriented incidence matrix for unsigned graphs, one can randomly choose an entry $\eta(i, ij)$ to be either +1 or -1, but the entry $\eta(j, ij)$ is then determined by $\sigma(ij)$ and so η is called an orientation of G_σ .

From here, we obtain two different yet virtually equivalent definitions for the line graph of a signed graph. The first is the combinatorial line graph $\mathcal{L}_C(\Sigma) = (L(G), \sigma_C)$, whose signature σ_C is determined by the adjacency matrix $A_{\mathcal{L}_C(\Sigma)}$ defined as follows:

$$A_{\mathcal{L}_C(\Sigma)} = 2I - B_\eta^T B_\eta.$$

The second is the spectral line graph $\mathcal{L}_S(\Sigma) = (L(G), \sigma_S)$ whose signature σ_S is determined by the adjacency matrix $A_{\mathcal{L}_S(\Sigma)}$ defined as follows:

$$A_{\mathcal{L}_S(\Sigma)} = B_\eta^T B_\eta - 2I.$$

The signed subdivision graph $\mathcal{S}$ of a signed graph $\Sigma = G_\sigma$ will have the underlying graph $S(G)$. As with the subdivision of a regular graph, this does not depend on the line graph but does depend on the vertex-edge incidence matrix. Since this is the same for both the combinatorial and spectral cases, we only need one definition to cover all signed graphs.

Definition 5.1. The signed subdivision graph $\mathcal{S}$ of $\Sigma = G_\sigma$ is the signed graph determined by

$$A_{\mathcal{S}(\Sigma)} = \begin{pmatrix} O & B_\eta \\ B_\eta^T & O \end{pmatrix}.$$

Similarly, the signed graph $\mathcal{Q}$ of $\Sigma = G_\sigma$ will have the underlying graph $Q(G)$. However, this time it does depend on the line graph and so different cases require consideration.

Definition 5.2. The graph $\mathcal{Q}$ of $\Sigma = G_\sigma$ is the signed graph determined by

$$A_{\mathcal{Q}_*(\Sigma)} = \begin{pmatrix} O & B_\eta \\ B_\eta^T & A_{\mathcal{L}_*(\Sigma_\eta)} \end{pmatrix},$$

where $* \in \{C, S\}$.

It is well known that two signed graphs $\Sigma = G_\sigma$ and $\Sigma' = G_{\sigma'}$ are switching equivalent if and only if there exists a $(-1, 0, 1)$ -diagonal matrix S called the switching matrix such that $A_{\Sigma'} = S^{-1}A_\Sigma S$. We say that the corresponding matrices are switching similar. We will show that our definitions of the graph $\mathcal{Q}_*(\Sigma)$ are stable under both reorientation and switching. For reorientation, we have the following lemma.

Lemma 5.3. Let $\Sigma = G_\sigma$ be a signed graph and Σ_η and $\Sigma_{\eta'}$ be two orientations of Σ . Then $\mathcal{Q}_*(\Sigma_\eta)$ and $\mathcal{Q}_*(\Sigma_{\eta'})$ are switching equivalent for each $* \in \{C, S\}$.

Proof. We will restrict the discussion to the spectral line graph. Hence hereafter $\mathcal{L}(\Sigma) := \mathcal{L}_S(\Sigma)$ and $\mathcal{Q}(\Sigma_\eta) := \mathcal{Q}_S(\Sigma_\eta)$.

Let $G = (V, E)$ where $|V| = n$ and $|E| = m$. Suppose that η and η' differ on some set $F \subseteq E$ and let B_η and $B_{\eta'}$ be the corresponding vertex-edge incidence matrices respectively. Let $S = (s_{ij})$ be the $m \times m$ diagonal matrix such that $s_{ii} = -1$ if $e_i \in F$ and $s_{ii} = 1$ otherwise. Then $B_{\eta'} = B_\eta S$. Since $S = S^T = S^{-1}$, we have

$$A_{\mathcal{L}(\Sigma_{\eta'})} = B_{\eta'}^T B_{\eta'} - 2I = S^T (B_\eta^T B_\eta - 2I) S = S^T A_{\mathcal{L}(\Sigma_\eta)} S.$$

Therefore,

$$\begin{aligned} A_{\mathcal{Q}(\Sigma_{\eta'})} &= \begin{pmatrix} O & B_{\eta'} \\ B_{\eta'}^T & A_{\mathcal{L}(\Sigma_{\eta'})} \end{pmatrix} \\ &= \begin{pmatrix} O & B_\eta S \\ S^T B_\eta^T & S^T A_{\mathcal{L}(\Sigma_\eta)} S \end{pmatrix} \\ &= \begin{pmatrix} I & O \\ O & S^T \end{pmatrix} \begin{pmatrix} O & B_\eta \\ B_\eta^T & A_{\mathcal{L}(\Sigma_\eta)} \end{pmatrix} \begin{pmatrix} I & 0 \\ 0 & S \end{pmatrix} \\ &= \begin{pmatrix} I & O \\ O & S \end{pmatrix}^{-1} A_{\mathcal{Q}(\Sigma_\eta)} \begin{pmatrix} I & O \\ O & S \end{pmatrix}. \end{aligned}$$

This completes the proof. $\square$

Now we will prove that any two switching equivalent signed graphs produce switching equivalent graphs under Q .

Lemma 5.4. *If Σ and Σ' are switching equivalent, then $Q_*(\Sigma)$ and $Q_*(\Sigma')$ are also switching equivalent for each $*$ $\in \{C, S\}$.*

Proof. The notation used in this proof shall be the same as in the previous lemma. Since Σ and Σ' are switching equivalent, their adjacency matrices are switching similar. Hence, $A_\Sigma = S^{-1}A_{\Sigma'}S$ for some switching matrix S . Observe that $B = B_\eta$ is a vertex-edge incidence matrix of Σ . Then $B' = SB_\eta$ is a vertex-edge incidence matrix of Σ' . Additionally, since here we are considering the spectral line graph, we have $A_{\mathcal{L}(\Sigma)} = B^TB - 2I$.

Therefore, for Σ' , we shall have

$$\begin{aligned} A_{Q(\Sigma')} &= \begin{pmatrix} O & B' \\ B'^T & A_{\mathcal{L}(\Sigma')} \end{pmatrix} \\ &= \begin{pmatrix} O & B' \\ B'^T & B'^TB' - 2I \end{pmatrix} \\ &= \begin{pmatrix} O & SB \\ (SB)^T & B^T(S^TS)B - 2I \end{pmatrix} \\ &= \begin{pmatrix} S & O \\ O & I \end{pmatrix} \begin{pmatrix} O & B \\ B^T & A_{\mathcal{L}(\Sigma)} \end{pmatrix} \begin{pmatrix} S^T & O \\ O & I \end{pmatrix} \\ &= \begin{pmatrix} S & O \\ O & I \end{pmatrix}^{-1} A_{Q(\Sigma)} \begin{pmatrix} S & O \\ O & I \end{pmatrix}. \end{aligned}$$

Hence, $Q(\Sigma)$ is switching equivalent to $Q(\Sigma')$ and we are done. $\square$

The signed graph $\mathcal{R}$ of $\Sigma = G_\sigma$ will have the underlying graph $R(G)$. The signature on the newly introduced connecting edges is given by the orientation η on Σ . For a vertex v of the root graph and an adjacent new vertex e corresponding to the edge e in the root graph, we have $\sigma_{\mathcal{R}}(v, e) = \eta(v, e)$. Note that this construction does not specify an orientation of the edge ve .

Like the signed subdivision graph the signed graph $\mathcal{R}(\Sigma)$ of $\Sigma = G_\sigma$ does not rely on either of the line graphs $\mathcal{L}_*(\Sigma)$, so a single definition will be sufficient.

Definition 5.5. The graph $\mathcal{R}$ of $\Sigma = G_\sigma$ is the signed graph determined by

$$A_{\mathcal{R}(\Sigma)} = \begin{pmatrix} A_\Sigma & B_\eta \\ B_\eta^T & O \end{pmatrix}.$$

The signed total graph $\mathcal{T}$ of $\Sigma = G_\sigma$ was defined and discussed in [9].

Definition 5.6. The total graph $\mathcal{T}$ of $\Sigma = G_\sigma$ is the signed graph determined by

$$A_{\mathcal{T}(\Sigma)} = \begin{pmatrix} A_\Sigma & B_\eta \\ B_\eta^T & A_{\mathcal{L}_*(\Sigma_\eta)} \end{pmatrix},$$

where $*$ $\in \{C, S\}$. For examples of these graphs see Figure 8.

5.3 Classes $(\mathcal{NC}_3^\Omega)$ and $(\mathcal{NC}_3^{\hat{\Omega}})$

5.3.1 Class $(\mathcal{NC}_3^\Omega)$ ($\Omega = S, Q, R, T$)

Suppose that G is an r -regular graph of order N and size M . Let $K_S(G) = S(G)$, $K_Q(G) = Q(G)$, $K_R(G) = R(G)$, and $K_T(G) = T(G)$. Also let $G_\Omega = K_\Omega(G) \circ K_1$. In this class, we will assume that for each Ω every edge in the graph K_Ω is positive. Then for $\Omega = S, Q, R, T$, the graph

$$\Gamma = G[G_\Omega, H_1, H_2, \dots, H_n, \hat{H}_1, \hat{H}_2, \dots, \hat{H}_m, F_1, F_2, \dots, F_k] \in (\mathcal{NC}_3^\Omega)^-$$

is obtained from the graphs G_Ω , H_a ($a = 1, 2, \dots, n$), $\hat{H}_b$ ($b = 1, 2, \dots, m$), and F_c ($c = 1, 2, \dots, k$) by joining the newly introduced vertices of $K_\Omega(G)$ – i.e. those not present in the root graph G – with the vertices u_{as} and v'_{as} of each H_a for $s = 1, 2, \dots, p_a$ and with the vertices $\hat{u}_{bt}$ and $\hat{v}'_{bt}$ of each $\hat{H}_b$ for $t = 1, 2, \dots, q_b$ and then also joining the old vertices of $K_\Omega(G)$ – i.e. those which are present in the root graph G – with all vertices of each and every F_c . Note that the order N and size M of the graph G are not reliant on n and m and k here.

Example 5.7. Let $\Gamma_4 = G[G_S, H_1, \hat{H}_1, F_1] \in (\mathcal{NC}_3^S)^-$ with $G_S = S(C_4) \circ K_1$, $H_1 = C_4 \circ K_1$, $\hat{H}_1 = \hat{C}_6 \circ K_1$, and $F_1 = C_4 \circ K_1$. This graph is shown in Figure 9. When factorised the characteristic polynomial of Γ_4 is

$$(x^2 - 1)^7(x^2 + 2x - 1)(x^4 - 4x^2 + 1)^2(x^4 - 5x^2 + 1)^2 \\ (x^{12} - 2x^{11} - 86x^{10} + 106x^9 + 1855x^8 - 532x^7 - 6612x^6 + 532x^5 + 1855x^4 - 106x^3 - 86x^2 + 2x + 1).$$

Hence, by Definition 2.9, the graph Γ_4 has the strong anti-reciprocal property (-SR).

Remark 5.8. To make some of the following computations simpler, rather than using vertex-edge incidence matrices we will use edge-vertex incidence matrices. This gives the relevant adjacency matrices for S, Q, L , and R slightly different forms to those discussed previously in this section. Hence, the adjacency matrix of $K_\Omega(G)$ is given by

$$A(K_\Omega(G)) = \begin{pmatrix} A_\Omega & B \\ B^T & C_\Omega \end{pmatrix},$$

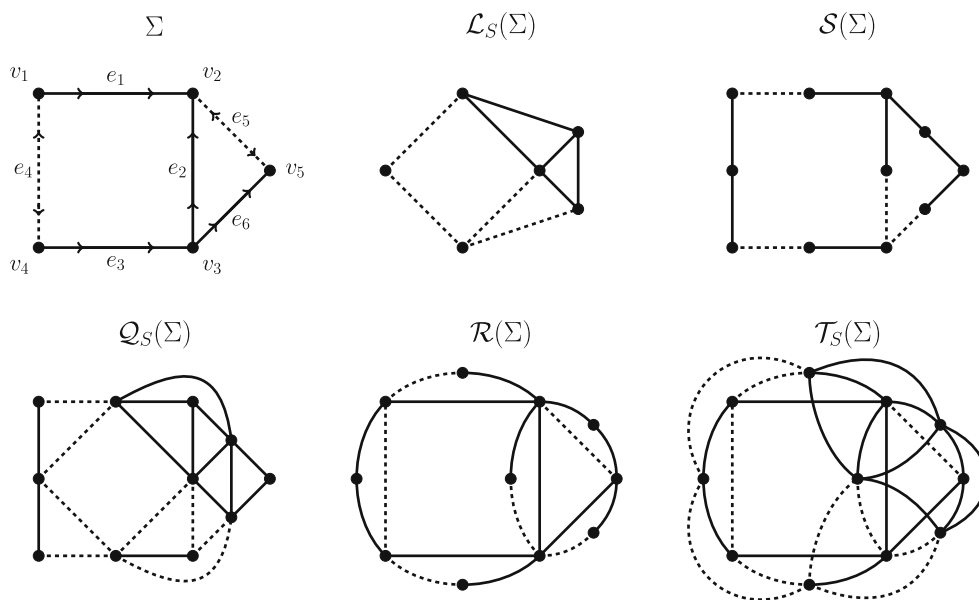


Figure 8: A signed graph Σ with an orientation along with its resulting spectral line graph $\mathcal{L}_S(\Sigma)$, its signed subdivision graph $\mathcal{S}(\Sigma)$, the related signed graphs $\mathcal{Q}_S(\Sigma)$ and $\mathcal{R}(\Sigma)$, and its spectral total graph $\mathcal{T}_S(\Sigma)$.

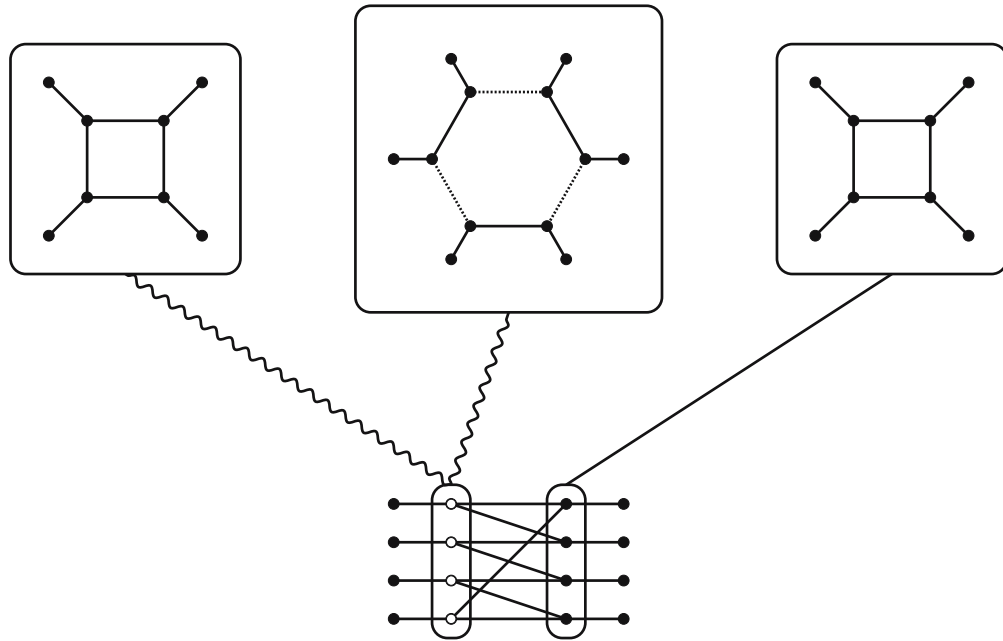


Figure 9: This is the graph $\Gamma_4 = G[G_S, H_1, \hat{H}_1, F_1] \in (\mathcal{NC}_3^S)$ with $G_S = S(C_4) \circ K_1$, $H_1 = C_4 \circ K_1$, $\hat{H}_1 = \hat{C}_6 \circ K_1$, and $F_1 = C_4 \circ K_1$. The edges between G_S and $H_1, \hat{H}_1$ will each connect to all four of the white vertices in $S(C_4)$ and the edges between G_S and F_1 will each connect to all four of the black vertices in $S(C_4)$. Unfortunately, this is difficult to draw without making little bit of a mess.

where A_Ω is a square matrix of order M such that $A_\Omega = 0$ for $\Omega = S, R$ and $A_\Omega = A(L(G))$ the adjacency matrix of the line graph of G for $\Omega = Q, T$ and C_Ω is a square matrix of order N such that $C_\Omega = 0$ for $\Omega = S, Q$ and $C_\Omega = A(G)$ for $\Omega = R, T$ and B is the $M \times N$ edge-vertex incidence matrix of G .

Let $\Omega' = 0$ for $\Omega = S, R$ and $\Omega' = 1$ for $\Omega = Q, T$. Also let $\Omega'' = 0$ for $\Omega = S, Q$ and $\Omega'' = 1$ for $\Omega = R, T$. Define B' to be the $M \times N$ rectangular diagonal matrix with diagonal entries

$$b_{ii} = \begin{cases} \sqrt{2r} & i = 1, \\ \sqrt{\lambda_i + r} & i = 2, 3, \dots, N, \\ 0 & i > N. \end{cases}$$

Since G is regular we can see from the singular value decomposition of the matrix B that the matrix $\begin{pmatrix} A_\Omega + aJ_N & B \\ B^T & C_\Omega + \beta J_M \end{pmatrix}$ is similar to the matrix $\begin{pmatrix} A'_\Omega & B' \\ B'^T & C'_\Omega \end{pmatrix}$, where A'_Ω is the diagonal matrix of order M with diagonal entries

$$a_{ii} = \begin{cases} \Omega'(2r - 2) + aM & i = 1, \\ \Omega'(\lambda_i + r - 2) & i = 2, 3, \dots, N, \\ 0 & i > N. \end{cases}$$

and C'_Ω is the diagonal matrix of order N with diagonal entries

$$c_{ii} = \begin{cases} \Omega''r + \beta N & i = 1, \\ \Omega''\lambda_i & i = 2, 3, \dots, N. \end{cases}$$

Like before, we can let $P = 4(p_1 + p_2 + \dots + p_n)$ and $Q = 4(q_1 + q_2 + \dots + q_m)$ and $L = 4(l_1 + l_2 + \dots + l_k)$. Then let $k_1 = P + Q$ and $k_2 = L$. By proper vertex labelling of a graph $\Gamma \in (\mathcal{NC}_3^0)^-$, its adjacency matrix can be written as follows:

$$A(\Gamma) = \begin{pmatrix} H & O_{k_1 \times k_2} & E & O_{k_1 \times N} & O_{k_1 \times M} & O_{k_1 \times N} \\ O_{k_2 \times k_1} & F & O_{k_2 \times M} & J_{k_2 \times N} & O_{k_2 \times M} & O_{k_2 \times N} \\ E^T & O_{M \times k_2} & A_\Omega & B & I_M & O_{M \times N} \\ O_{N \times k_1} & J_{N \times k_2} & B^T & C_\Omega & O_{N \times M} & I_N \\ O_{M \times k_1} & O_{M \times k_2} & I_M & O_{M \times N} & O_M & O_{M \times N} \\ O_{N \times k_1} & O_{N \times k_2} & O_{N \times M} & I_N & O_{N \times M} & O_N \end{pmatrix},$$

where H and F and E are the same as we defined prior to Theorem 4.2.

Theorem 5.9. *The graph $\Gamma \in (\mathcal{NC}_3^{\Omega})^-$ satisfies the property (-SR).*

Proof. The characteristic polynomial of the adjacency matrix of $\Gamma \in (\mathcal{NC}_3^{\Omega})^-$ is

$$\det(xI - A(\Gamma)) = \det \begin{pmatrix} xI_{k_1} - H & O_{k_1 \times k_2} & -E & O_{k_1 \times N} & O_{k_1 \times M} & O_{k_1 \times N} \\ O_{k_2 \times k_1} & xI_{k_2} - F & O_{k_2 \times M} & -J_{k_2 \times N} & O_{k_2 \times M} & O_{k_2 \times N} \\ -E^T & O_{M \times k_2} & xI_M - A_\Omega & -B & -I_M & O_{M \times N} \\ O_{N \times k_1} & -J_{N \times k_2} & -B^T & xI_N - C_\Omega & O_{N \times M} & -I_N \\ O_{M \times k_1} & O_{M \times k_2} & -I_M & O_{M \times N} & xI_M & O_{M \times N} \\ O_{N \times k_1} & O_{N \times k_2} & O_{N \times M} & -I_N & O_{N \times M} & xI_N \end{pmatrix}.$$

By applying Schur's formula as in Lemma 2.5 to this determinant, we obtain $\det(xI - A(\Gamma))$

$$\begin{aligned} &= \det \begin{pmatrix} xI_{k_1} - H & O_{k_1 \times k_2} \\ O_{k_2 \times k_1} & xI_{k_2} - F \end{pmatrix} \times \det \begin{pmatrix} xI_M - A_\Omega & -B & -I_M & O_{M \times N} \\ -B^T & xI_N - C_\Omega & O_{N \times M} & -I_N \\ -I_M & O_{M \times N} & xI_M & O_{M \times N} \\ O_{N \times M} & -I_N & O_{N \times M} & xI_N \end{pmatrix} \\ &\quad - \begin{pmatrix} -E^T & O_{M \times k_2} \\ O_{N \times k_1} & -J_{N \times k_2} \end{pmatrix} \begin{pmatrix} xI_{k_1} - H & O_{k_1 \times k_2} \\ O_{k_2 \times k_1} & xI_{k_2} - F \end{pmatrix}^{-1} \begin{pmatrix} -E & O_{k_1 \times N} & O_{k_1 \times M} & O_{k_1 \times N} \\ O_{k_2 \times M} & -J_{k_2 \times N} & O_{k_2 \times M} & O_{k_2 \times N} \end{pmatrix} \\ &= \det((xI_{k_1} - H)(xI_{k_2} - F)) \det \begin{pmatrix} xI_M - A_\Omega - \alpha J_M & -B & -I_M & O_{M \times N} \\ -B^T & xI_N - C_\Omega - \beta J_N & O_{N \times M} & -I_N \\ -I_M & O_{M \times N} & xI_M & O_{M \times N} \\ O_{N \times M} & -I_N & O_{N \times M} & xI_N \end{pmatrix}, \end{aligned}$$

where

$$\alpha = E^T(xI_{k_1} - H)^{-1}E = \sum_{a=1}^n E_{p_a}^T(xI_{4p_a} - A(H_a))^{-1}E_{p_a} + \sum_{b=1}^m E_{q_b}^T(xI_{4q_b} - A(\hat{H}_b))^{-1}E_{q_b},$$

$$\beta = J_{N \times k_2}(xI_{k_2} - F)^{-1}J_{k_2 \times N} = \sum_{c=1}^k \mathbf{1}_{4l_c}^T(xI_{4l_c} - A(F_c))^{-1}\mathbf{1}_{4l_c}.$$

Now by Lemmas 2.11 and 2.13, we obtain

$$\alpha = \frac{Px(x^2 - 1)}{2(x^4 - 6x^2 + 1)} + \frac{Qx}{2(x^2 - 1)} \quad \text{and} \quad \beta = \frac{Lx}{x^2 - 2x - 1}.$$

Reapplying Schur's formula gives us that $\det(xI - A(\Gamma))$

$$\begin{aligned}
&= \det((xI_{k_1} - H)(xI_{k_2} - F)) \det(xI_M) \det(xI_N) \\
&\quad \times \det \left(\begin{pmatrix} xI_M - A_\Omega - aJ_M & -B \\ -B^T & xI_N - C_\Omega - \beta J_N \end{pmatrix} - (-I_{M+N})(xI_{M+N})^{-1}(-I_{M+N}) \right) \\
&= \det((xI_{k_1} - H)(xI_{k_2} - F)) \times x^{N+M} \\
&\quad \times \det \left(\begin{pmatrix} \left(x - \frac{1}{x}\right)I_M - A_\Omega - aJ_M & -B \\ -B^T & \left(x - \frac{1}{x}\right)I_N - C_\Omega - \beta J_N \end{pmatrix} \right).
\end{aligned}$$

From here, we can make use of Remark 5.8 and then once again apply Schur's complement formula to obtain to $\det(xI - A(\Gamma))$

$$\begin{aligned}
&= \det((xI_{k_1} - H)(xI_{k_2} - F)) \times x^{N+M} \times \det \left(\begin{pmatrix} \left(x - \frac{1}{x}\right)I_M - A'_\Omega & -B' \\ -B'^T & \left(x - \frac{1}{x}\right)I_N - C'_\Omega \end{pmatrix} \right) \\
&= \det((xI_{k_1} - H)(xI_{k_2} - F)) \times x^{N+M} \times \det \left(\left(x - \frac{1}{x}\right)I_N - C'_\Omega \right) \\
&\quad \times \det \left(\left(x - \frac{1}{x}\right)I_M - A'_\Omega - B' \left(\left(x - \frac{1}{x}\right)I_N - C'_\Omega \right)^{-1} B'^T \right).
\end{aligned}$$

Case I: $\Omega = S$. For $\Omega = S$, we have $\Omega' = 0$ and $\Omega'' = 0$. Hence, the characteristic polynomial $\det(xI - A(\Gamma))$ shall be

$$\begin{aligned}
&= \det((xI_{k_1} - H)(xI_{k_2} - F)) \times x^{N+M} \times \det \left(\left(x - \frac{1}{x}\right)I_N - \text{diag}(\beta N, 0, \dots, 0) \right) \\
&\quad \times \det \left(\left(x - \frac{1}{x}\right)I_M - \text{diag}(aM, 0, \dots, 0) - B' \left(\left(x - \frac{1}{x}\right)I_N - \text{diag}(\beta N, 0, \dots, 0) \right)^{-1} B'^T \right) \\
&= \prod_{a=1}^n P(H_a; x) \prod_{b=1}^m P(\hat{H}_b; x) \prod_{c=1}^k P(F_c; x) \times x^{N+M} \times \left(x - \frac{1}{x} - \beta N\right) \left(x - \frac{1}{x}\right)^{N-1} \\
&\quad \times \left(x - \frac{1}{x} - aM - \frac{2r}{x - \frac{1}{x} - \beta N}\right) \times \prod_{i=2}^N \left(x - \frac{1}{x} - \frac{\lambda_i + r}{x - \frac{1}{x}}\right) \times \left(x - \frac{1}{x}\right)^{M-N} \\
&= \prod_{a=1}^n P(H_a; x) \prod_{b=1}^m P(\hat{H}_b; x) \prod_{c=1}^k P(F_c; x) \prod_{i=2}^N \left(x - \frac{1}{x}\right) \prod_{i=2}^N \left(x - \frac{1}{x} - \frac{\lambda_i + r}{x - \frac{1}{x}}\right) \\
&\quad \times x^{N+M} \left(x - \frac{1}{x}\right)^{M-N} \left(x - \frac{1}{x} - \beta N\right) \left(x - \frac{1}{x} - aM - \frac{2r}{x - \frac{1}{x} - \beta N}\right) \\
&= \prod_{a=1}^n P(H_a; x) \prod_{b=1}^m P(\hat{H}_b; x) \prod_{c=1}^k P(F_c; x) \prod_{i=2}^N (x^4 - (\lambda_i + r + 2)x^2 + 1) \times \left(\frac{1}{x^2}\right)^{N-1} \\
&\quad \times x^{N+M} \left(x - \frac{1}{x}\right)^{M-N} \left(x - \frac{1}{x} - \beta N\right) \left(x - \frac{1}{x} - aM - \frac{2r}{x - \frac{1}{x} - \beta N}\right) \\
&= \prod_{a=1}^n P(H_a; x) \prod_{b=1}^m P(\hat{H}_b; x) \prod_{c=1}^k P(F_c; x) \prod_{i=2}^N (x^4 - (\lambda_i + r + 2)x^2 + 1) \\
&\quad \times (x^2 - 1)^{M-N} x^2 \left(x - \frac{1}{x} - \beta N\right) \left(x - \frac{1}{x} - aM - \frac{2r}{x - \frac{1}{x} - \beta N}\right).
\end{aligned}$$

After plugging in our expressions for α and β and doing a little rearranging, we will have the final form of the characteristic polynomial of the graph $\Gamma \in (\mathcal{NC}_3^S)^-$

$$P(\Gamma; x) = \frac{(x^2 - 1)^{M-N} \Psi_S(x)}{2x^8 - 4x^7 - 16x^6 + 28x^5 + 28x^4 - 28x^3 - 16x^2 + 4x - 2} \\ \times \prod_{a=1}^n P(H_a; x) \prod_{b=1}^m P(\hat{H}_b; x) \prod_{c=1}^k P(F_c; x) \prod_{i=2}^N (x^4 - (\lambda_i + r + 2)x^2 + 1),$$

where $\Psi_S(x)$

$$= 2x^{12} - 4x^{11} - (PM + QM + 2LN + 4r + 20)x^{10} \\ + (2PM + 2QM + 8r + 36)x^9 + (PLMN + QLMN + 4PM + 8QM + 16LN + 32r + 62)x^8 \\ - (6PM + 14QM + 56r + 88)x^7 - (2PLMN + 6QLMN + 6PM + 14QM + 28LN + 56r - 88)x^6 \\ + (6PM + 14QM + 56r + 88)x^5 + (PLMN + QLMN + 4PM + 8QM + 16LN + 32r + 62)x^4 \\ - (2PM + 2QM + 8r + 36)x^3 - (PM + QM + 2LN + 4r + 20)x^2 + 4x + 2.$$

Now the result for $\Omega = S$ follows by Definition 2.9.

Case II: $\Omega = Q$. For $\Omega = Q$, we have $\Omega' = 1$ and $\Omega'' = 0$. Following similar steps as in Case I, we will find the characteristic polynomial of the graph $\Gamma \in (\mathcal{NC}_3^Q)^-$

$$P(\Gamma; x) = \frac{(x^2 - 1)^{M-N} \Psi_Q(x)}{2x^8 - 4x^7 - 16x^6 + 28x^5 + 28x^4 - 28x^3 - 16x^2 + 4x - 2} \\ \times \prod_{a=1}^n P(H_a; x) \prod_{b=1}^m P(\hat{H}_b; x) \prod_{c=1}^k P(F_c; x) \\ \times \prod_{i=2}^N (x^4 - (\lambda_i + r - 2)x^3 - (\lambda_i + r + 2)x^2 + (\lambda_i + r - 2)x + 1),$$

where $\Psi_Q(x)$

$$= 2x^{12} - 4rx^{11} + (PM + QM - 2LN + 4r - 28)x^{10} + (2PM + 2QM + 4LN - 4LNr - 44r)x^9 \\ - (PLMN + QLMN + 4PM + 8QM - 16LN + 32r - 126)x^8 + (6PM + 14QM + 28LN - 28LNr - 144r)x^7 \\ + (2PLMN + 6QLMN + 6PM + 14QM - 28LN + 56r - 200)x^6 - (6PM + 14QM + 28LN - 28LNr - 144r)x^5 \\ - (PLMN + QLMN + 4PM + 8QM - 16LN + 32r - 126)x^4 - (2PM + 2QM + 4LN - 4LNr - 44r)x^3 \\ + (PM + QM - 2LN + 4r - 28)x^2 + 4rx + 2.$$

Now the result for $\Omega = Q$ follows by Definition 2.9.

Case III: $\Omega = R$. For $\Omega = R$, we have $\Omega' = 0$ and $\Omega'' = 1$. Following similar steps as in Case I we will find the characteristic polynomial of the graph $\Gamma \in (\mathcal{NC}_3^R)^-$

$$P(\Gamma; x) = \frac{(x^2 - 1)^{M-N} \Psi_R(x)}{2x^8 - 4x^7 - 16x^6 + 28x^5 + 28x^4 - 28x^3 - 16x^2 + 4x - 2} \\ \times \prod_{a=1}^n P(H_a; x) \prod_{b=1}^m P(\hat{H}_b; x) \prod_{c=1}^k P(F_c; x) \\ \times \prod_{i=2}^N (x^4 - \lambda_i x^3 - (\lambda_i + r + 2)x^2 + \lambda_i x + 1),$$

where $\Psi_R(x)$

$$\begin{aligned}
&= 2x^{12} - (2r + 4)x^{11} - (PM + QM + 2LN + 20)x^{10} + (2PM + 2QM + PMr + QMr + 26r + 36)x^9 \\
&\quad + (PLMN + QLMN + 4PM + 8QM + 16LN - 2PMr - 2QMr + 62)x^8 \\
&\quad - (6PM + 14QM + 3PMr + 7QMr + 100r + 88)x^7 \\
&\quad - (2PLMN + 6QLMN + 6PM + 14QM + 28LN - 4PMr - 12QMr + 88)x^6 \\
&\quad + (6PM + 14QM + 3PMr + 7QMr + 100r + 88)x^5 \\
&\quad + (PLMN + QLMN + 4PM + 8QM + 16LN - 2PMr - 2QMr + 62)x^4 \\
&\quad - (2PM + 2QM + PMr + QMr + 26r + 36)x^3 \\
&\quad - (PM + QM + 2LN + 20)x^2 + (2r + 4)x + 2.
\end{aligned}$$

Now the result for $\Omega = R$ follows by Definition 2.9.

Case IV: $\Omega = T$. For $\Omega = T$, we have $\Omega' = 1$ and $\Omega'' = 1$. Following similar steps as in Case I, we will find the characteristic polynomial of the graph $\Gamma \in (\mathcal{NC}_3^T)^-$

$$\begin{aligned}
P(\Gamma; x) &= \frac{(x^2 - 1)^{M-N} \Psi_T(x)}{2x^8 - 4x^7 - 16x^6 + 28x^5 + 28x^4 - 28x^3 - 16x^2 + 4x - 2} \\
&\quad \times \prod_{a=1}^n P(H_a; x) \prod_{b=1}^m P(\hat{H}_b; x) \prod_{c=1}^k P(F_c; x) \\
&\quad \times \prod_{i=2}^N (x^4 - (2\lambda_i + r - 2)x^3 - (\lambda_i^2 + r\lambda_i - 3\lambda_i - r - 2)x^2 + (2\lambda_i + r - 2)x + 1),
\end{aligned}$$

where $\Psi_T(x)$

$$\begin{aligned}
&= 2x^{12} - 6rx^{11} - (PM + QM + 2LN - 4r^2 - 4r + 28)x^{10} \\
&\quad + (2PM + 2QM - 4LN + PMr + QMr + 4LNr - 8r^2 + 70r)x^9 \\
&\quad + (PLMN + QLMN + 4PM + 8QM + 16LN - 2PMr - 2QMr - 32r^2 - 32r + 126)x^8 \\
&\quad - (6PM + 14QM - 28LN + 3PMr + 7QMr + 28LNr - 56r^2 + 244r)x^7 \\
&\quad - (2PLMN + 6QLMN + 6PM + 14QM + 28LN - 4PMr - 12QMr - 56r^2 - 56r + 200)x^6 \\
&\quad + (6PM + 14QM - 28LN + 3PMr + 7QMr + 28LNr - 56r^2 + 244r)x^5 \\
&\quad + (PLMN + QLMN + 4PM + 8QM + 16LN - 2PMr - 2QMr - 32r^2 - 32r + 126)x^4 \\
&\quad - (2PM + 2QM - 4LN + PMr + QMr + 4LNr - 8r^2 + 70r)x^3 \\
&\quad - (PM + QM + 2LN - 4r^2 - 4r + 28)x^2 + 6rx + 2.
\end{aligned}$$

Now the result for $\Omega = T$ follows by Definition 2.9. □

5.3.2 Class $(\mathcal{NC}_3^{\hat{\Omega}})$ ($\hat{\Omega} = S, Q_S, R, T_S$)

Suppose that $\Sigma = G_\sigma$ is an r -regular signed graph of order N and size M . Let $K_S(\Sigma) = S(\Sigma)$, $K_{Q_S}(\Sigma) = Q_S(\Sigma)$, $K_R(\Sigma) = R(\Sigma)$, and $K_{T_S}(\Sigma) = T_S(\Sigma)$. Also let $\Sigma_{\hat{\Omega}} = K_{\hat{\Omega}}(\Sigma) \circ K_1$. Then for $\hat{\Omega} = S, Q_S, R, T_S$, the graph

$$\Gamma = G[\Sigma_{\hat{\Omega}}, H_1, H_2, \dots, H_n, \hat{H}_1, \hat{H}_2, \dots, \hat{H}_m, F_1, F_2, \dots, F_k] \in (\mathcal{NC}_3^{\hat{\Omega}})^-$$

is obtained from the graphs $\Sigma_{\hat{\Omega}}$, H_a ($a = 1, 2, \dots, n$), $\hat{H}_b$ ($b = 1, 2, \dots, m$), and F_c ($c = 1, 2, \dots, k$) by joining the newly introduced vertices of $K_{\hat{\Omega}}(\Sigma)$ – i.e. those not present in the root graph Σ – with the vertices u_{as} and v'_{as} of each H_a for $s = 1, 2, \dots, p_a$ and with the vertices $\hat{u}_{bt}$ and $\hat{v}'_{bt}$ of each $\hat{H}_b$ for $t = 1, 2, \dots, q_b$ and then also joining the old vertices of $K_{\hat{\Omega}}(\Sigma)$ – i.e. those which are present in the root graph Σ – with all vertices of each and every F_c . Note that the order N and size M of the graph Σ are not reliant on n and m and k here.

We can see here that this time the graph which connects to the corona cycles will in general not have an all-positive signature. Following a similar method as in the proof of Theorem 5.9, we obtain the following theorem.

Theorem 5.10. *The graph $\Gamma \in (\mathcal{NC}_3^0)^-$ satisfies the property (-SR).*

6 Concluding remarks

In this article, we have taken to the realm of signed graphs the search for graphs with the properties (SR) and (-SR). It is interesting to observe how the results obtained for unsigned graphs can be ported to signed graphs with suitable settings.

It is easy to see that when making the alternating join between a bipartite signed graph and corona cycles of even length, we retain bipartiteness in the compound graph. So, for the classes in this article, in the cases where only the alternating join is present and the root graph is bipartite, the properties (SR) and (-SR) are equivalent.

We want to emphasise the fact that in many parts of the constructions we request net-regularity because this is necessary to achieve the all-one eigenvector. However, we can see that such a constraint is not mandatory for the graphs to possess the properties (SR) and (-SR), but it is necessary to easily derive the complete polynomial. If we remove the net-regularity conditions, then the eigenvalues from (1) in the first class of graphs $(\mathcal{NC}_1^1)^-$ will be a larger family and the computation becomes more involved. We leave this as an open problem for further study.

Open Problem 6.1. Prove the property (-SR) for the constructions of this article and of [14] to signed graphs in which the (net-)regularity is no longer requested.

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References

- [1] U. Ahmad, S. Hameed, and S. Jabeen, *Class of weighted graphs with strong anti-reciprocal eigenvalue property*, Linear Multilinear Algebra **68** (2020), no. 6, 1129–1139.
- [2] S. Akbari, F. Belardo, E. Dodongeh, and M. A. Nematollahi, *Spectral characterizations of signed cycles*, Linear Algebra Appl. **553** (2018), 307–327.
- [3] S. Akhter, S. Hameed, and U. Ahmad, *Signed graphs with strong anti-reciprocal eigenvalue property*, Commun. Algebra **51** (2023), 4271–4279.

- [4] R. B. Bapat, *Graphs and Matrices*, 2nd edition, Springer, London, 2014.
- [5] S. Barik, S. Ghosh, and D. Mondal, *On graphs with strong anti-reciprocal eigenvalue property*, Linear Multilinear Algebra **70** (2022), no. 21, 6698–6711.
- [6] S. Barik, M. Nath, S. Pati, and B. K. Sarma, *Unicyclic graphs with strong reciprocal eigenvalue property*, Electron. J. Linear Algebra **17** (2008), 139–153.
- [7] S. Barik, S. Pati, and B. K. Sarma, *The spectrum of the corona of two graphs*, SIAM J. Discrete Math. **21** (2007), no. 1, 47–56.
- [8] F. Belardo, S. Cioaba, J. Koolen, and J. F. Wang, *Open problems on the spectral theory of signed graphs*, Art Discrete Appl. Math. **1** (2018), no. 2, #P2.10.
- [9] F. Belardo, Z. Stanić, and T. Zaslavsky, *Total graph of a signed graph*, Ars Math. Contemp. **23** (2023), #P1.02, 1–17.
- [10] D. M. Cvetković, M. Doob, and H. Sachs, *Spectra of Graphs: Theory and Applications*, 3rd edition, J. A. Barth Verlag, Leipzig, 1995.
- [11] E. Ghorbani, W. H. Haemers, H. R. Maimani, and L. P. Majd, *On sign-symmetric signed graphs*, Ars Math. Contemp. **19** (2020), no. 1, 83–83.
- [12] S. K. Panda and S. Pati, *Graphs with reciprocal eigenvalue properties*, Electron. J. Linear Algebra **31** (2016), 511–514.
- [13] S. Pirzada, T. Shamsheer, and M. A. Bhat, *A note on the class of weighted graphs with strong anti-reciprocal eigenvalue property*, Linear Multilinear Algebra **71** (2023), 1–4.
- [14] B. R. Rakshith, K. C. Das, and B. J. Manjunatha, *Some new families of noncorona graphs with strong anti-reciprocal eigenvalue property*, Bull. Malays. Math. Sci. Soc. **45** (2022), 2597–2618.
- [15] P. M. Weichsel, *The Kronecker product of graphs*, Proc. Amer. Math. Soc. **18** (1962), 47–52.
- [16] T. Zaslavsky, *Signed graphs*, Discrete Appl. Math. **4** (1982), 47–74.